数理分析方法

向量、凸性与渐近锥

Lecture 109

原课件逐页保留 · 零基础中文讲解 · 证明训练

材料与编写说明

课程原页：张顺明教授课件（用户提供）。参考：Gérard Debreu, Theory of Value (1959)，第一章 Mathematics。原页与本讲义的补充、纠错、例题分开呈现。本文为辅助自学材料，不代表教师官方讲义。

怎样阅读

先看本页的中文说明，再对照原页符号；遇到证明，先抄出假设和目标，再逐行说明“这一步用的是什么”。遮住证明重写一遍，最后用反例检查条件是否必要。重复或动画展开页仍逐页保留；说明会指出新增步骤或对应的完整论证位置。

“补充”用于补基础或展开原课件略去的论证；“纠错”指出原页需要修正的地方。课程概览及人物介绍页按其原有用途解释，不虚构数学定理。

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零基础读证明：随手查阅

对象的层次。 $x \in A$ 表示一个元素属于集合，$A \subseteq B$ 表示 A 的每个元素都在 B 中。$\{x\}$ 是只含 x 的集合，不是 x 本身。$\varnothing$ 是空集，$\{\varnothing\}$ 则含有一个元素。$\mathbb{R}^N$ 中的点是有 N 个坐标的有序列表。

量词。 $\forall$ 是“对所有”，$\exists$ 是“至少存在一个”。例如“每个学生有一本书”不要求是同一本；“有一本书供每个学生使用”则要求同一本。证明极限时 $\forall \varepsilon > 0\ \exists n_0\ \forall n \geq n_0$ 的顺序不可换：先指定精度，再找统一起点，最后控制这个起点之后的全部项。

怎样否定。 否定“所有 x 满足 P”得到“存在 x 不满足 P”；否定“存在”得到“所有都不”。否定 $P \Rightarrow Q$ 是 P 真且 Q 假。因此要推翻一个定理，必须满足其全部假设又违反结论；拿不满足假设的例子不能当反例。

四种基本证明。 (1) 包含 $A \subseteq B$：任取 $x \in A$，用定义得到 $x \in B$。(2) 相等 $A = B$：两个方向分别证明。(3) 等价 $P \iff Q$：证明 $P \Rightarrow Q$ 及反向。(4) 反证：把结论否定加入已知，推出矛盾。逆否命题 $\neg Q \Rightarrow \neg P$ 与 $P \Rightarrow Q$ 等价，“反过来” $Q \Rightarrow P$ 则未必成立。

数列、子列和收敛。 数列 x_n 是第 n 个对象；子列 x_{n_k} 的指标必须严格递增。$x_n \to x$ 指任意小的误差最终都能保持。前面有限多项不影响极限。抽子列通常是为了用有界性和有限维 Bolzano–Weierstrass 定理获得极限；抽取之后须把原来所有相关量一起限制到同一子列。

距离的关键估计。 $\|x\| = (\sum_i x_i^2)^{1/2}$，$\|x - y\|$ 是距离。三角不等式 $\|x - z\| \leq \|x - y\| + \|y - z\|$ 允许经中间点拆误差。两段误差各小于 $\varepsilon/2$，总误差便小于 ε；分成 m 段时可各分配 ε/m。这里的 ε 不是固定的小数，而是任意正精度。

做证明题的五行草稿。 先写完整假设；写准确目标；列出可用定义与定理；逐步注明所用条件；最后检查任意性、存在性和量词顺序。出现“显然”时问：若阅卷者要求展开，我能否补出一个集合等价链、一个不等式或一个明确构造？不知道证明的命题要标为引用，不能把待证结论循环当作理由。

学习导航：向量、凸性与无穷远方向

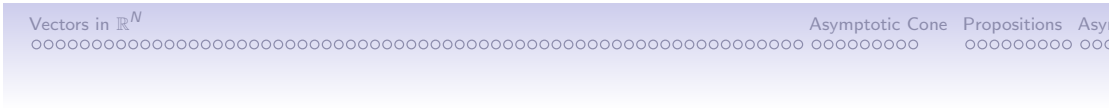
本讲对应 Debreu《Theory of Value》第一章第9节。课件后半部分补充了渐近锥理论，并重复前半部分若干证明。原页全部保留；重复页仍逐页说明其作用，并指向首次完整论证。先掌握第3–23页的运算，再学第46–66页的凸性，最后回到渐近锥，是较适合初学者的阅读顺序。

先认符号。 $\mathbb{R}^N$ 是所有 N 个实数组成的有序数组；$x=(x_1,\dots,x_N)$，$\|x\|=(\sum_i x_i^2)^{1/2}$ 是到原点的距离。$x_k\to x$ 表示距离 $\|x_k-x\|\to 0$。下标 k 常是序列编号，分量编号另用 i，不要混为一谈。$\overline{C}$ 表示闭包，即 C 内点列所有可能的极限组成的集合；$\operatorname{int}C$ 表示能在 C 中放下一个小开球的点。

本讲反复使用的四件工具。 三角不等式 $\|u+v\|\le\|u\|+\|v\|$；有限维有界序列有收敛子列；闭集包含其内部点列的全部极限；在 $\mathbb{R}^N$ 中紧集恰为闭且有界的集合。注意“有界”只保证某个子列收敛，绝不保证原序列收敛。写证明时必须保留“取子列”这一步。

渐近锥的约定。 本文将课件 AC、C_∞ 统一理解为非空集合的渐近锥，且总含 0；0^+C 专指衰退锥。它们对非空闭凸集相等，对一般集合不相等。遇到空集合的交、和时先单独处理，避免套用要求非空的序列定义。

1 原课件第 1 页：本讲研究什么



Theory of Value

An Axiomatic Analysis of Economic Equilibrium

Gerard Debreu (1959)

Professor Shunming Zhang

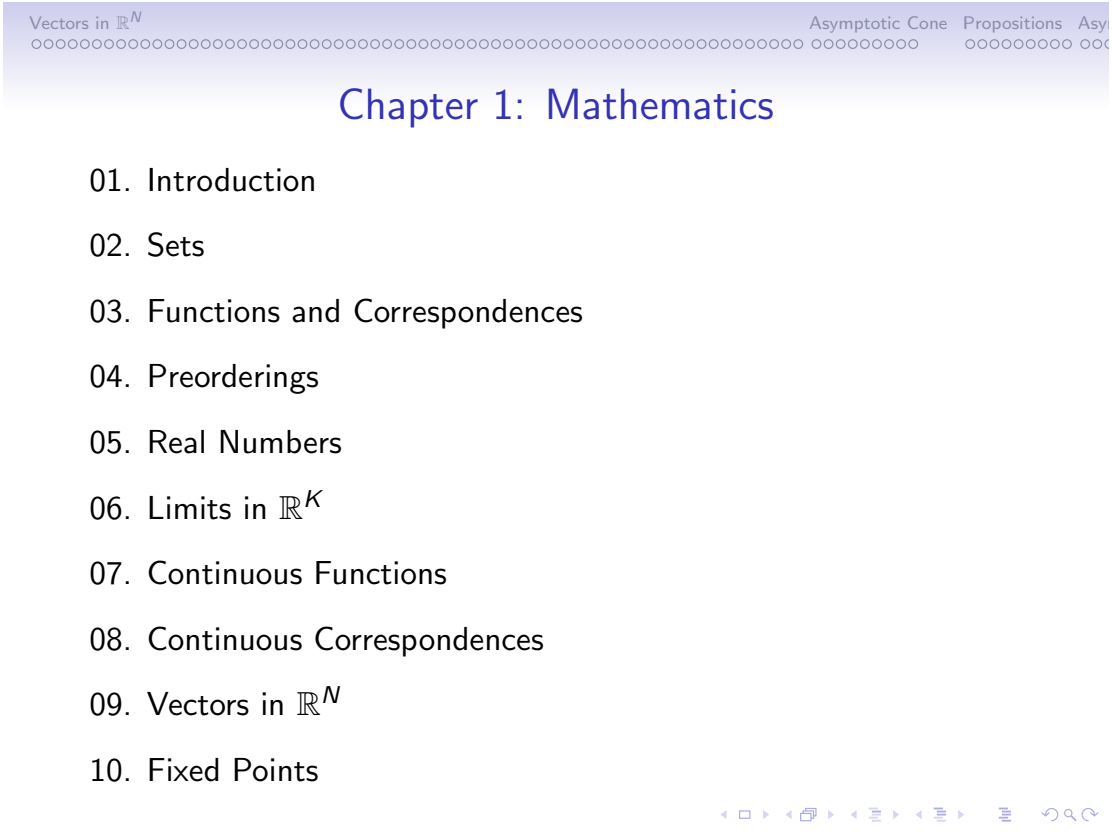
School of Finance
Renmin University of China

October 1, 2022

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本讲标题是“$\mathbb{R}^N$ 中的向量”，但内容远超向量计算：研究集合相加是否保持闭性、凸性，以及无界集合往哪些方向延伸。经济学中多个生产集合相加就是总生产可能集合；证明这个总集合闭，才能保证极限方案仍可行。封面给出教师、教材与授课时间，不含需要证明的命题。

2 原课件第 2 页：在课程知识链中的位置



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

前面的极限、连续性和对应理论，本讲都要用到。主线是：向量加法连续 $\Rightarrow$ 紧集之和紧；无穷远方向不能互相抵消 $\Rightarrow$ 闭集之和闭；凸性 $\Rightarrow$ 可用超平面描述边界。下一讲的不动点定理又需要紧性与凸性。本页的目录不是一组定理，先用它定位依赖关系。

3 原课件第 3 页：向量相加：逐个坐标相加

Vectors in $\mathbb{R}^N$

a. Let $x^1 = \left(x_n^1\right)$ and $x^2 = \left(x_n^2\right)$ be two elements of $\mathbb{R}^N$; one defines their sum $x^1 + x^2$ as $\left(x_n^1 + x_n^2\right)$, i.e., the nth coordinate of $x^1 + x^2$ is the sum of the nth coordinates of x^1 and x^2. The element 0 has been defined (1.6.d) by the condition that all its coordinates are equal to 0. The negative of x is $-x = (-x_n)$. One writes $x^1 - x^2$ for $x^1 + \left(-x^2\right)$. If $x^1, \cdots, x^K, \cdots, x^K$ are K elements of $\mathbb{R}^N$, their sum is denoted $\sum_{n=1}^{K} x^K$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $x^1 = (1, 2)$、$x^2 = (3, -1)$，则 $x^1 + x^2 = (4, 1)$，$-x^1 = (-1, -2)$，$x^1 - x^2 = (-2, 3)$。上标 $1, 2$ 标记两个向量，并非幂。一般写 $\sum_{k=1}^{K} x^k$，其第 i 个坐标是 $\sum_{k=1}^{K} x_i^k$。课件末行的求和指标排版有混用，记住“k 数向量、i 数坐标”即可。零向量的每个坐标都是 0。

4 原课件第 4 页：数乘及平行四边形直观



Vectors in $\mathbb{R}^N$

b. Let $x = (x_n)$ be an element of $\mathbb{R}^N$, and t a real number; one defines their product tx, or xt, as (tx_n). Geometrically, given two points x^1 and x^2 of $\mathbb{R}^N$, one obtains $x^1 + x^2$ by completing the parallelogram having $0x^1$, $0x^2$ for sides; one obtains $-x$ by taking the symmetric of x with respect to 0: one obtains tx by representing $\mathbb{R}$ an the straight line $0x$, choosing 0 to represent 0, x to represent 1; then $t \in \mathbb{R}$ is represented by tx.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$tx = (tx_1, \ldots, tx_N)$：$t > 1$ 拉长，$0 < t < 1$ 缩短，$t < 0$ 还反向，$t = 0$ 变为原点。向量加法可看成先走 x^1 再走 x^2；终点是平行四边形的对角顶点。若 $x = 0$，所谓“直线 $0x$”不唯一，但代数定义仍给 $t0 = 0$，不受影响。

5 原课件第 5 页：为什么加法和数乘连续



c. It is clear that the functions $(x^1, x^2) \to x^1 + x^2$ from $\mathbb{R}^N \times \mathbb{R}^N$ to $\mathbb{R}^N$, and $(t, x) \to tx$ from $\mathbb{R} \times \mathbb{R}^N$ to $\mathbb{R}^N$ are continuous.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

加法的完整验证。 若 $u_k \to u$、$v_k \to v$，则

$$\|(u_k + v_k) - (u + v)\| \le \|u_k - u\| + \|v_k - v\| \longrightarrow 0.$$

数乘的完整验证。 若 $t_k \to t$、$x_k \to x$，把差拆成 $t_k(x_k - x) + (t_k - t)x$。因为收敛数列 (t_k) 有界，存在 M 使 $|t_k| \le M$，于是

$$\|t_k x_k - tx\| \le M\,\|x_k - x\| + |t_k - t|\,\|x\| \longrightarrow 0.$$

这是课件“显然连续”背后的理由。考试不能只说“极限可以交换”，应给出连续性或以上估计。

6 原课件第 6 页：点与向量是两种看法



d. The elements of $\mathbb{R}^N$ are also called vectors.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

同一个数组 $(x_1, \ldots, x_N)$，说“点”时强调位置，说“向量”时强调从原点到该点的位移。比如 $x - y$ 是从 y 指向 x 的位移。后面的集合平移和法向量都用这种转换；不需要另学一种新的对象。

7 原课件第 7 页：平移



Vectors in $\mathbb{R}^N$

e. Given a vector a of $\mathbb{R}^N$, the transformation of $\mathbb{R}^N$ into itself defined by $x \to x + a$ is called the *a*-translation in $\mathbb{R}^N$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

固定 a，$T_a(x) = x + a$ 将所有点沿同一方向搬动。它连续，逆映射 T_{-a} 也连续，因此保留开集、闭集、紧集、连通性。证明闭性也可直接做：若 $x_k + a \to z$ 且 $x_k \in C$，则 $x_k \to z - a$；C 闭推出 $z - a \in C$，故 $z \in C + a$。平移不改变无穷远方向，见第31页。

8 原课件第 8 页：集合相加：闵可夫斯基和



f. Let X^1 and X^2 be two subsets of $\mathbb{R}^N$; one defines their sum $X^1 + X^2$ as the set of elements of $\mathbb{R}^N$ of the form $x^1 + x^2$ where $x^1 \in X^1$ and $x^2 \in X^2$. In other words, one takes, in all possible ways, an element of X^1 and an element of X^2 and adds them; the set of elements thus obtained is $X^1 + X^2$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$A+B=\{a+b : a\in A, b\in B\}$，意为两边各挑一个点，再把所有可能的和收集起来。它与并集 $A\cup B$ 完全不同。例如 $[0,1]+[2,4]=[2,5]$，而并集是两段。证明 $z\in A+B$ 时，应展示分解 $z=a+b$；证明不存在这种分解即可证明不属于和集。

9 原课件第 9 页：两条线段相加得到正方形

Vectors in $\mathbb{R}^N$

Example: (1) Let $X^1 = \{x \in \mathbb{R}^2 \mid 0 \leqq x_1 \leqq 1, x_2 = 0\}$, and $X^2 = \{x \in \mathbb{R}^2 \mid x_1 = 0, 0 \leqq x_2 \leqq 1\}$. Then $X^1 + X^2 = \{x \in \mathbb{R}^2 \mid 0 \leqq x_1 \leqq 1, 0 \leqq x_2 \leqq 1\}$. (2) See also Figure 4 in $\mathbb{R}^2$.

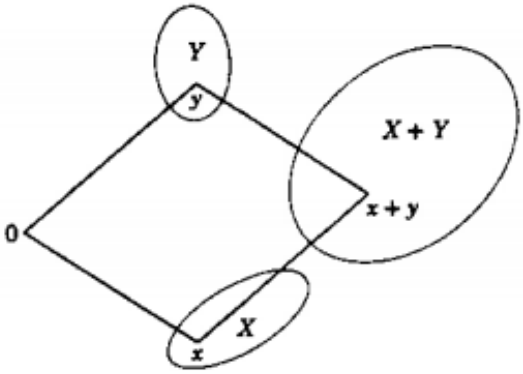


Fig. 4

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

课件中的第一条线段是 $A = \{(s, 0) : 0 \leq s \leq 1\}$，第二条是 $B = \{(0, t) : 0 \leq t \leq 1\}$。任取两点相加得到 $(s, t) \in [0, 1]^2$，故 $A + B \subset [0, 1]^2$；反过来每个 (s, t) 都可写为 $(s, 0) + (0, t)$，故另一包含也成立。集合等式通常要这样证明两个方向，不能只凭图形判断。

10 原课件第 10 页：集合取负、作差与闭包的和

Vectors in $\mathbb{R}^N$

One defines $-X$ as the set of elements of $\mathbb{R}^N$ of the form $-x$ where $x \in X$. One writes $X^1 - X^2$ for $X^1 + \left(-X^2\right)$. If $X_1, \cdots, X_k, \cdots, X_K$ are K subsets of $\mathbb{R}^N$, their sum is denoted $\sum_{k=1}^{K} X_k$. It is easy to prove :

(1) If $X_1, \cdots, X_k, \cdots, X_K$ are K subsets of $\mathbb{R}^N$, then $\sum_{k=1}^{K} \overline{X_k} \subseteq \overline{\sum_{k=1}^{K} X_k}$.

That is, the sum of their adherences is contained in the adherence of their sum.

Proof. If $X_1, \cdots, X_k, \cdots, X_K$ are K subsets of $\mathbb{R}^N$, then for any $k = 1, \cdots, K$ and any $x_k \in \overline{X_k}$, there exists $\left\{x_k^q; q \in \mathbb{N}\right\} \subseteq X_k$ such that $\lim_{q\to\infty} x_k^q = x_k$. Then $\left\{\sum_{k=1}^{K} x_k^q; q \in \mathbb{N}\right\} \subseteq \overline{\sum_{k=1}^{K} X_k}$ and $\lim_{q\to\infty} \sum_{k=1}^{K} x_k^q = \sum_{k=1}^{K} x_k \in \overline{\sum_{k=1}^{K} X_k}$, hence $\sum_{k=1}^{K} \overline{X_k} \subseteq \overline{\sum_{k=1}^{K} X_k}$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$-A = \{-a : a \in A\}$，$A - B = A + (-B)$，并非集合差 $A \setminus B$。本页命题正确读作

$$\overline{A_1} + \cdots + \overline{A_K} \subseteq \overline{A_1 + \cdots + A_K}.$$

任取左边的 $z = \sum_k a_k$。因为 $a_k \in \overline{A_k}$，对每个 k 选 $a_{k,q} \in A_k$ 使 $a_{k,q} \to a_k$。有限个极限可以相加，故 $z_q = \sum_k a_{k,q} \to z$，且 z_q 在和集内。按闭包定义，z 在右边。横线所在位置至关重要；这并没有说闭集之和一定闭。

11 原课件第 11 页：紧集之和紧：先看有界性

Vectors in $\mathbb{R}^N$

One can also prove:

(2) The sum of K compact subsets of $\mathbb{R}^N$ is compact.

Proof. In $\mathbb{R}^N$, a set is compact if and only if it is closed and bounded.

If $X_1, \cdots, X_k, \cdots, X_K$ are K compact subsets of $\mathbb{R}^N$, then they are bounded, there exists a constant M_k such that $\|x_k\| \leqq M_k$ for $x_k \in X_k$, $k = 1, \cdots, K$. For the sum $\sum_{k=1}^{K} X_k$, any element x can be written as $x = \sum_{k=1}^{K} x_k$, where $x_k \in X_k$, thus

$$\|x\| = \left|\sum_{k=1}^{K} x_k\right| \leqq \sum_{k=1}^{K} |x_k| \leqq \sum_{k=1}^{K} M_k.$$

Therefore, the sum $\sum_{k=1}^{K} X_k$ is bounded.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 A_k 紧，则有常数 M_k 使 $\|a_k\| \le M_k$。任取 $z \in \sum_k A_k$，有 $z = \sum_k a_k$，从而 $\|z\| \le \sum_k \|a_k\| \le \sum_k M_k$。上界不依赖所选 z，故整个和集有界。如果有一个 $A_k = \varnothing$，和集为空，紧性自动成立；下面默认各集合非空。此页只完成“有界”，还不能宣告紧。

12 原课件第 12 页：紧集之和紧：连续像的短证明

Vectors in $\mathbb{R}^N$

One can also prove:

(2) The sum of K compact subsets of $\mathbb{R}^N$ is compact.

Proof. The sum of closed sets is not necessarily closed. However, in $\mathbb{R}^N$, the sum of compact (which are closed and bounded) sets is closed. This is because the sum of compact sets can be viewed as the image of the compact set $\prod_{k=1}^{K} X_k$ under the continuous function $(x_1, \cdots, x_K) \mapsto \sum_{k=1}^{K} x_k$. Since the continuous image of a compact set is compact, the sum $\sum_{k=1}^{K} X_k$ is closed.

Since the sum of K compact subsets of $\mathbb{R}^N$ is both bounded and closed, it is compact.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

有限乘积 $P = A_1 \times \cdots \times A_K$ 紧，映射 $F(a_1, \ldots, a_K) = \sum_k a_k$ 连续，而 $F(P) = \sum_k A_k$。连续映射把紧集送到紧集，所以结论成立。这条论证其实已同时得到闭性和有界性。

为什么不能把紧换成闭。 在 $\mathbb{R}^2$ 中取 $A = \{(u, 1/u) : u > 0\}$、$B = \{(v, 0) : v \in \mathbb{R}\}$，两者都闭，然而 $A + B = \mathbb{R} \times (0, \infty)$ 不闭。A 的闭性来自：若它的点列有有限极限，则第一坐标不可能趋于 0 或无穷，极限仍满足乘积为 1 且正。

13 原课件第 13 页：紧集之和紧：逐层抽取子列

Vectors in $\mathbb{R}^N$

One can also prove:

(2) The sum of K compact subsets of $\mathbb{R}^N$ is compact.

Proof. If $X_1, \cdots, X_k, \cdots, X_K$ are K compact subsets of $\mathbb{R}^N$, for any sequence $\{x^q; q \in \mathbb{N}\} \subseteq \sum_{k=1}^{K} X_k$, we assume $x^q = \sum_{k=1}^{K} x_k^q$ such that $x_k^q \in X_k$ for all $k = 1, \cdots, K$. For compact set X_1, there exists a subsequence $\{x_1^{q_1}; q_1 \in \mathbb{N}\}$ of sequence $\{x_1^q; q \in \mathbb{N}\}$ such that $\lim_{q_1 \to \infty} x_1^{q_1} = x_1^0 \in X_1$. For compact set X_2, there exists a subsequence $\{x_2^{q_2}; q_2 \in \mathbb{N}\}$ of sequence $\{x_2^{q_1}; q_1 \in \mathbb{N}\}$ such that $\lim_{q_2 \to \infty} x_2^{q_2} = x_2^0 \in X_2$ with $\lim_{q_2 \to \infty} x_1^{q_2} = x_1^0 \in X_1$. Continuing this procedure, we have that, for compact set X_K, there exists a subsequence $\{x_K^{q_K}; q_K \in \mathbb{N}\}$ of sequence $\{x_K^{q_{K-1}}; q_{K-1} \in \mathbb{N}\}$ such that $\lim_{q_K \to \infty} x_K^{q_K} = x_K^0 \in X_K$ with $\lim_{k_K \to \infty} x_k^{q_K} = x_k^0 \in X_k$ for $k = 1, \cdots, K-1$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

不用连续像定理也能证明。给定 $z_q = \sum_{k=1}^{K} a_{k,q}$，先利用 A_1 紧，挑索引子列，使第一个分量收敛；只在这条子列中再挑，使第二个分量收敛。第一个分量仍然收敛，因为收敛序列的子列保留原极限。重复有限次，得到同一条索引子列 q_j，使每个 $a_{k,q_j} \to a_k \in A_k$。不能分别随意选 K 条不同索引子列后直接相加。

14 原课件第 14 页：紧集之和紧：把子列重新相加

Vectors in $\mathbb{R}^N$

Therefore

$$\lim_{q_K\to\infty} x_k^{q_K} = x_k^0 \in X_k, \qquad \text{for} \quad k = 1, \cdots, K$$

hence

$$\lim_{q_K\to\infty} \sum_{k=1}^{K} x_k^{q_K} = \sum_{k=1}^{K} x_k^0 \in \sum_{k=1}^{K} X_k$$

that is, there exists a subsequence $\{x^{q_K}; q_K \in \mathbb{N}\}$ of sequence $\{x^q; q \in \mathbb{N}\}$ such that

$$\lim_{q_K\to\infty} x^{q_K} = \lim_{q_K\to\infty} \sum_{k=1}^{K} x_k^{q_K} = \sum_{k=1}^{K} x_k^0 = x^0 \in \sum_{k=1}^{K} X_k$$

Thus, $\sum_{k=1}^{K} X_k$ are compact. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

沿第13页得到的共同索引，

$$z_{q_j} = \sum_{k=1}^{K} a_{k,q_j} \longrightarrow \sum_{k=1}^{K} a_k \in \sum_{k=1}^{K} A_k.$$

于是和集的每个序列都有极限仍在和集内的收敛子列，正是有限维空间中的序列紧性。这页收尾的逻辑顺序是“任取序列—构造子列—验证极限留在集合”，不是“找到了某个收敛序列所以紧”。

15 原课件第 15 页：有限个连续函数相加仍连续



g. Let S be a subset of $\mathbb{R}^N$, and for every $k = 1, \cdots, K$, let f_k be a function from S to $\mathbb{R}^L$. Define a function f from S to $\mathbb{R}^L$ by $f(x) = \sum_{k=1}^{K} f_k(x)$ for every x in S. It is immediate that:

(3) If every f_k is continuous at the point x of S, then f is continuous at x.

(3′) The sum of K continuous functions at the point x of S is continuous at x.

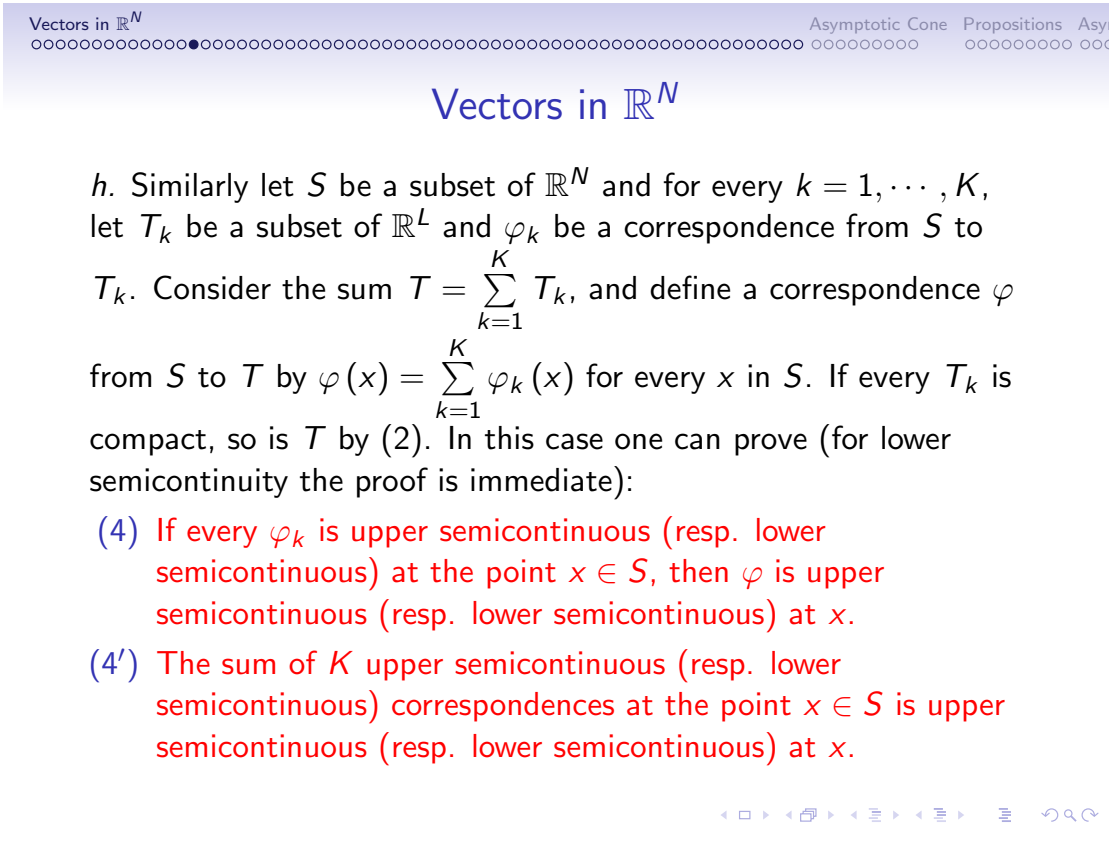
上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 $f = \sum_{k=1}^{K} f_k$，要证在 x_0 连续。给定 $\varepsilon > 0$，每个 f_k 连续，选 $\delta_k > 0$ 使 $\|x - x_0\| < \delta_k$ 时 $\|f_k(x) - f_k(x_0)\| < \varepsilon/K$。取 $\delta = \min_k \delta_k > 0$，三角不等式给

$$\|f(x) - f(x_0)\| \le \sum_k \|f_k(x) - f_k(x_0)\| < \varepsilon.$$

“有限”保证最小值仍为正；无限级数不能直接照抄，通常要增加一致收敛等条件。

16 原课件第 16 页：对应相加：为什么需要控制值域



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

对应 $\varphi_k(x)$ 给出一个可选集合；$\varphi(x) = \sum_k \varphi_k(x)$ 给出所有总选择。课件把各值域放在紧集 T_k 中，以便从分解的各分量抽取收敛子列。采用本教材的紧值、闭图式上半连续约定时，上半连续性得以保持。若使用一般拓扑教材中不自动要求闭值的“开集定义”，须另外明确紧值等条件，不能不加条件地把闭图与上半连续等同。下半连续表示旧选择能由邻近参数下的选择逼近；求和的下半连续不需要各 T_k 紧。

17 原课件第 17 页：对应之和半连续：补全漏掉的抽子列步骤

Vectors in $\mathbb{R}^N$

Proof. If $x^q \to x^0$, $y^q \in \varphi(x^q)$ and $y^q \to y^0$, then $y^q = \sum_{k=1}^{K} y_k^q$ with $y_k^q \in \varphi_k(x^q)$ and $y^0 = \sum_{k=1}^{K} y_k^0$ with $y_k^q \to y_k^0$. If every φ_k is upper semicontinuous at the point $x^0 \in S$, then $y_k^0 \in \varphi_k(x^0)$. Thus

$$y^0 = \sum_{k=1}^{K} y_k^0 \in \sum_{k=1}^{K} \varphi_k\left(x^0\right) = \varphi\left(x^0\right)$$

φ is upper semicontinuous at x^0.

If $x^q \to x^0$ and $y^0 \in \varphi(x^0)$, then $y^0 = \sum_{k=1}^{K} y_k^0$ with $y_k^0 \in \varphi_k(x^0)$. If every φ_k is lower semicontinuous at the point $x^0 \in S$, then there is (y_k^q) such that $y_k^q \in \varphi_k(x^q)$ and $y_k^q \to y_k^0$. Thus

$$y^q = \sum_{k=1}^{K} y_k^q \in \sum_{k=1}^{K} \varphi_k(x^q) = \varphi(x^q), \qquad y^q = \sum_{k=1}^{K} y_k^q \to \sum_{k=1}^{K} y_k^0 = y^0$$

φ is lower semicontinuous at x^0. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

上半连续。 设 $x_q \to x_0$，$y_q \in \varphi(x_q)$，$y_q \to y$。分解 $y_q = \sum_k y_{k,q}$，其中 $y_{k,q} \in \varphi_k(x_q) \subset T_k$。由各 T_k 紧，取共同子列使 $y_{k,q_j} \to y_k$。各对应在 x_0 的闭图性质给 $y_k \in \varphi_k(x_0)$，故 $y = \sum_k y_k \in \varphi(x_0)$。总值域紧，因此这推出所需上半连续性。原页直接写各分量收敛，省略了关键依据。

下半连续。 给定 $y = \sum_k y_k \in \varphi(x_0)$ 和任意 $x_q \to x_0$。各 φ_k 下半连续，存在 $y_{k,q} \in \varphi_k(x_q)$ 且 $y_{k,q} \to y_k$；取和便得 $y_q \in \varphi(x_q)$ 且 $y_q \to y$。这里要逼近的是指定的旧选择 y。

18 原课件第 18 页：加权平均与凸组合的区别



i. Let x^1, x^2 be two points of $\mathbb{R}^N$, t^1, t^2 two real numbers such that $t^1 + t^2 = 1$. The point $t^1x^1 + t^2x^2$ is called the weighted average of x^1 and x^2 with weights t^1 and t^2 (respectively).

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$t_1 + t_2 = 1$ 时 $t_1x^1 + t_2x^2$ 称仿射组合，课件称 weighted average。若再有 $t_1, t_2 \geq 0$，才是凸组合。比如 $2x^1 - x^2$ 的系数和为 1，却可能跑到线段外。后面证明凸性时必须检查“系数非负”和“系数和为 1”两个条件。

19 原课件第 19 页：直线、射线：参数的范围决定形状

Vectors in $\mathbb{R}^N$

j. Let x, y be two different points of $\mathbb{R}^N$.

The straight line x, y is $\left\{z \in \mathbb{R}^N \mid t \in \mathbb{R}, z = (1-t)\,x + ty\right\}$.

The closed half-line x, y (the origin x is written first) is $\left\{z \in \mathbb{R}^N \mid t \in \mathbb{R}, 0 \leqq t, z = (1-t)\,x + ty\right\}$.

The open half-line x, y is $\left\{z \in \mathbb{R}^N \mid t \in \mathbb{R}, 0 < t, z = (1-t)\,x + ty\right\}$.

It may be more suggestive to rewrite the expression of z as $z = x + t\,(y - x)$, i.e., one multiplies the fixed vector $(y - x)$ by the variable number t; this gives the straight line (or the half-line) 0, $(y - x)$; then one adds the fixed vector x (which amounts to a translation; see fig. 5 drawn for the case of the half-line).

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

统一公式是 $z = x + t(y - x)$。$t \in \mathbb{R}$ 给整条直线；$t \geq 0$ 给从 x 出发经过 y 的闭射线；$t > 0$ 去掉起点，给开射线。取 $x = (1, 1)$、$y = (3, 2)$，则 $z = (1 + 2t, 1 + t)$。当 $t = 0, 1, 2$ 时分别是起点、y、继续向前的点。“闭射线”指包含端点，并不是有界。

20 原课件第 20 页：图5：方向相同，起点不同

Vectors in $\mathbb{R}^N$

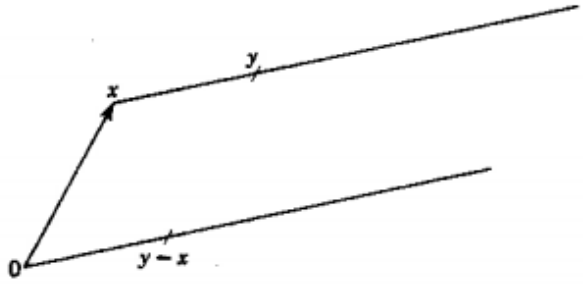


Fig. 5

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

图中下方射线从 0 出发，方向是 $y-x$；上方射线由下方整条平移 x 得到，起点变成 x 并经过 y。因此“方向”不包含起点信息。图里的斜率在二维画面可看见，但 $\mathbb{R}^N$ 中更可靠的描述始终是 $x+t(y-x)$。

21 原课件第 21 页：线段与退化线段



k. Let x_1, x_2 be two points of $\mathbb{R}^N$ (different or not).
The closed segment x_1, x_2, denoted $[x_1, x_2]$, is
$\{z \in \mathbb{R}^N \mid t \in \mathbb{R}, 0 \leqq t \leqq 1, z = (1-t)\,x_1 + tx_2\}$. (See fig. 5.)
When $x_1 = x_2$, the closed segment $[x_1, x_2]$ is said to be degenerate.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

把第19页的参数限制为 $0 \leq t \leq 1$，得到 $[x^1, x^2]$。两端都包括在内。若 $x^1 = x^2$，任意 t 都给同一个点，线段退化成单点。以后用“包含任意两点的整段”定义凸性时，单点也应允许，不能强行要求两点不同。

22 原课件第 22 页：锥：沿射线可以一直走

Vectors in $\mathbb{R}^N$

I. Given a subset C of $\mathbb{R}^N$ and a point x in C, C is said to be a cone with vertex x if it contains the closed half-line x, y whenever it owns the point y (different from x).

"$y \in C$" implies "$x + t\,(y - x) \in C$ for $0 \leqq t$."

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

顶点为 x 的锥 C 要满足：$x \in C$，且任意 $y \in C$、$t \geq 0$ 都有 $x + t(y - x) \in C$。顶点为 0 时简化为 $ty \in C$。锥不一定凸：二维正横轴与正纵轴的并是锥，却不包含 $(1,0)$ 与 $(0,1)$ 的中点。凸锥还要容许不同方向的非负线性组合。

23 原课件第 23 页：正半独立：方向不能抵消

Vectors in $\mathbb{R}^N$

m. Given K cones $C_1, \cdots, C_k, \cdots, C_K$ with vertex 0, they are said to be positively semi-independent if "$x_k \in C_k$ for every k, and $\sum_{k=1}^{K} x_k = 0$" implies "$x_k = 0$ for every k," i.e., if it is impossible to take a vector in each cone so that their sum is 0, unless they are all equal to 0. It is clear that two cones C_1, C_2 with vertex 0 are positively semi-independent if and only if $C_1 \cap (-C_2) = \{0\}$.

Proof. If $C_1 \cap (-C_2) = \{0\}$ and for $x_1 \in C_1$ and $x_2 \in C_2$ with $x_1 + x_2 = 0$, then $x_1 = -x_2 \in C_1 \cap (-C_2) = \{0\}$, and hence $x_1 = x_2 = 0$. Thus, C_1 and C_2 are positively semi-independent.

If C_1 and C_2 with vertex 0 are positively semi-independent, and if $x \in C_1 \cap (-C_2)$, then $x \in C_1$ and $-x \in C_2$ with $x + (-x) = 0$, then $x = 0$. Hence $C_1 \cap (-C_2) = \{0\}$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

锥 $C_1, \ldots, C_K$ 正半独立，意思是 $x_k \in C_k$ 且 $\sum_k x_k = 0$ 只能发生在全部 $x_k = 0$ 时。两个锥时等价于 $C_1 \cap (-C_2) = \{0\}$。

两方向证明。 若交仅含 0，由 $x_1 + x_2 = 0$ 得 $x_1 = -x_2 \in C_1 \cap (-C_2)$，所以两者均零。反过来，若正半独立，任取 $z \in C_1 \cap (-C_2)$，则 $z \in C_1$、$-z \in C_2$，两者和为零，故 $z = 0$。两个同向非负射线正半独立，两个相反射线则不满足。

24 原课件第 24 页：渐近锥：只保留越来越远的点

Vectors in $\mathbb{R}^N$

n. Consider a subset S of $\mathbb{R}^N$. To describe those of its points that are infinitely far from 0 one introduces the concept of its asymptotic cone, as follow. Let M be a non-negative real number, and denote by S^M the set $\{x \in S \mid \|x\| \geqq M\}$ of vectors in S whose norm is greater than or equal to M. Let $\Gamma\left(S^M\right)$ be the least closed cone with vertex 0 containing S^M (i.e., the intersection of all the closed cones with vertex 0 containing S^M). The asymptotic cone of S, denoted $\mathbf{A}S$, is defined as the intersection of all the $\Gamma\left(S^M\right)$, i.e.,

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right);$$

it is clearly a closed cone with vertex 0.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

先截取 $S^M = \{x \in S : \|x\| \geq M\}$，再作包含它的最小闭锥 $\Gamma(S^M)$，最后取所有 $M \geq 0$ 的交。这样的方向必须在无论多远的截尾中都能被逼近。锥在这里总含原点，特别约定 $\Gamma(\varnothing) = \{0\}$。每个 $\Gamma(S^M)$ 闭且对非负数乘封闭，交仍闭且是锥，所以 AS 是闭锥。它不一定凸，除非再对 S 加凸性条件。

25 原课件第 25 页：为什么截尾锥递减；闭包不能漏



Vectors in $\mathbb{R}^N$

Remark. If $M_1 \leqq M_2$, then $S^{M_1} \supseteq S^{M_2}$ and $\Gamma\left(S^{M_1}\right) \supseteq \Gamma\left(S^{M_2}\right)$. $\Gamma\left(S^M\right)$ is a monotonically decreasing set-sequence. Hence

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right) = \lim_{M \to \infty} \Gamma\left(S^M\right).$$

Therefore $\mathbf{A}S$ is the cone generated by the points in S which are at infinity.

As $\Gamma\left(S^M\right)$ is the least closed cone with vertex 0 containing S_M, then we can see that

$$\begin{aligned} \Gamma\left(S^M\right) &= \overline{\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\}} \\ &= \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S^M}\right\} \end{aligned}$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $M_1 \leq M_2$，距离门槛提高，故 $S^{M_2} \subset S^{M_1}$，进而 $\Gamma(S^{M_2}) \subset \Gamma(S^{M_1})$。所谓极限就是递减集合族的交。准确的公式是

$$\Gamma(E) = \overline{\{\alpha x : \alpha \geq 0,\ x \in E\} \cup \{0\}}.$$

闭包负责加入仅能逼近的边界方向；$\{0\}$ 负责空集情形。去掉闭包、或把“存在任意接近的点”误写成“恰好存在同方向的点”，都会使后面的证明失效。

26 原课件第 26 页：序列定义：把远点缩小后取极限

Vectors in $\mathbb{R}^N$

Alfred Auslender and Marc Teboulle (2003) *Asymptotic Cones and Functions in Optimization and Variational Inequalities*

Definition 1.2. Let C be a non-empty set in $\mathbb{R}^N$. Then the asymptotic cone of the set C, denoted by C_∞, is the set of vectors $d \in \mathbb{R}^N$ that are limits in direction of the sequences $\{x_k; k \in \mathbb{N}\} \subseteq C$, namely

$$C_\infty = \Big\{ d \in \mathbb{R}^N \mid \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq C \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \Big\}$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

对非空 C，$d \in C_\infty$ 等价于存在 $x_k \in C$、$t_k \to +\infty$ 使 $x_k/t_k \to d$。这里可以删掉有限个初项，使 $t_k > 0$。例：$C = \{(s,1) : s \geq 0\}$ 的渐近锥为 $\{(a,0) : a \geq 0\}$，因为缩放后第二坐标 $1/t_k \to 0$。零方向总存在：固定一点 $x \in C$，用 $x/k \to 0$。非零方向才表示真正逃向无穷远。

27 原课件第 27 页：衰退锥：从每个点都能沿它走

Vectors in $\mathbb{R}^N$

R. Tyrrell Rockafellar (1970) *Convex Analysis*

Definition 1.3. Let C be a non-empty convex set in $\mathbb{R}^N$.

- We shall say that C recedes in the direction D if C includes all the half-lines in the direction D which start at points of C. In other words, C recedes in the direction of y, where $y \neq 0$, if and only if $x + \lambda y \in C$ for every $\lambda \geqq 0$ and $x \in C$.
- The set of all vectors $y \in \mathbb{R}^N$ satisfying the latter condition, including $y = 0$, will be called the recession cone of C.
- The **recession cone** of C will be denoted by $\mathbf{0}^+C$.

$$\mathbf{0}^+C = \left\{ y \in \mathbb{R}^N \mid x + \lambda y \in C \text{ for all } x \in C \text{ and } \lambda \geqq 0 \right\}$$

- Directions in which C recedes will also be referred to as directions of recession of C.
- The recession cone of $\overline{C}$ has elsewhere been called the asymptotic cone of C.

$$\mathbf{A}C = \mathbf{0}^+\overline{C}.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

定义 $0^+C = \{d : x + \lambda d \in C \text{ 对所有 } x \in C, \lambda \geq 0\}$。这比“存在远点列沿该方向”强得多。一般只有 $0^+C \subset C_\infty$；C 非空闭凸时才可保证相等，完整证明见第110–111页。**条件提示：** 本页末行 $AC = 0^+C$ 若只假设凸而未假设闭，并不普遍成立；第112页正好给反例。不要把不同作者的术语相同误当作无条件等价。

28 原课件第 28 页：渐近锥性质总览与两个边界条件

Vectors in $\mathbb{R}^N$

Proposition 2.1. Let $C \subseteq \mathbb{R}^N$ be non-empty.

(1) C_∞ is a closed cone.

(2) $\left(\overline{C}\right)_\infty = C_\infty$.

(3) If C is a cone, then $C_\infty = \overline{C}$.

Theorem 2.1. If S is a subset of $\mathbb{R}^N$, then $\mathbf{A}S = S_\infty$.

Proposition 2.2. Let $C \subseteq \mathbb{R}^N$ be non-empty and define the normalized set

$$C_N = \left\{ d \in \mathbb{R}^N \mid \text{there exists } \{x_k; k \in \mathbb{N}\} \subseteq C \text{ with } \|x_k\| \to \infty \text{ such that } d = \lim_{k\to\infty} \frac{x_k}{\|x_k\|} \right\}$$

then $C_\infty = \mathrm{Cone}\,(C_N)$, where for any set C,
$\mathrm{Cone}\,(C_N) = \left\{\lambda x \in \mathbb{R}^N \mid x \in C_N, \lambda \geqq 0\right\}$.

Proposition 2.3. A set $C \subseteq \mathbb{R}^N$ is bounded if and only if $C_\infty = \{0\}$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

正确读法：C_∞ 闭且是锥；$(\overline{C})_\infty = C_\infty$；若 C 本身是锥，则 $C_\infty = \overline{C}$，横线不能漏。证明在第77–80页。 归一化方向集 C_N 是无界点列 x_k 的单位方向 $x_k/\|x_k\|$ 的极限集。为兼容有界 C 导致 $C_N = \varnothing$ 的情形，等式应写

$$C_\infty = \{0\} \cup \{\lambda u : \lambda > 0,\ u \in C_N\}.$$

“C 有界当且仅当 $C_\infty = \{0\}$”按本页上下文要求 C 非空，空集另行处理。

29 原课件第 29 页：有界当且仅当没有非零渐近方向

Vectors in $\mathbb{R}^N$

Proposition 2.3. A set $C \subseteq \mathbb{R}^N$ is bounded if and only if $C_\infty = \{0\}$.

Proof. It is clear that C_∞ cannot contain any non-zero direction if C is bounded. As C is bounded, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = 0$. Hence $C_\infty = \{0\}$.

Conversely, if C is unbounded, then there exists a sequence $\{x_k; k \in \mathbb{N}\} \subseteq C$ with $x_k \neq 0$ for all $k \in \mathbb{N}$ such that $\|x_k\| \to \infty$ and thus the vectors $d_k = \frac{x_k}{\|x_k\|} \in \{d \in \mathbb{R}^N : \|d\| = 1\}$.
Therefore, we can extract a subsequence of $\{d_k\}$ such that $\lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} d_k = d$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|d\| = 1$. This non-zero vector d is an element of C_∞, a contradiction! □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $\|x\| \le M$ 对所有 $x \in C$ 成立，那么任意 x_k/t_k 的范数至多 $M/t_k \to 0$，故只可能得到 0；非空性保证 0 确实存在。反向用逆否命题：若 C 无界，对每个整数 k 选 $x_k \in C$ 使 $\|x_k\| \ge k$。单位向量 $u_k = x_k/\|x_k\|$ 位于紧的单位球面，所以有子列趋向 u，且 $\|u\| = 1$。取 $t_j = \|x_{k_j}\| \to \infty$，就得到 $u \in C_\infty$，是非零方向。关键是“无界—归一化—球面紧—非零极限”。

30 原课件第 30 页：单调性、平移不变性与乘积的预告

Vectors in $\mathbb{R}^N$

o. It is obvious that "$S_1 \subseteq S_2$" implies "$\mathbf{A}S_1 \subseteq \mathbf{A}S_2$".

Proof. If $S_1 \subseteq S_2$, then $S_1^M \subseteq S_2^M$ for $M \geqq 0$, thus $\Gamma\left(S_1^M\right) \subseteq \Gamma\left(S_2^M\right)$ for $M \geqq 0$, and hence $\mathbf{A}S_1 \subseteq \mathbf{A}S_2$.

One can prove that:

(5) If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S + \{x\}) = \mathbf{A}S$.

In other words, a translation of S does not alter its asymptotic cone. It is then easy to prove that:

(6) If $T \neq \varnothing$ and S are two subsets of $\mathbb{R}^N$, then $\mathbf{A}S \subseteq \mathbf{A}(S + T)$.

One can also prove that:

(7) If, for every $k = 1, \cdots, K$, S_k is a subset of $\mathbb{R}^{N_k}$, then $\mathbf{A}\left(\prod_{k=1}^{K} S_k\right) \subseteq \prod_{k=1}^{K} \mathbf{A}S_k$.

Proof of (6). Let $x \in T$. Then $\mathbf{A}S = \mathbf{A}(S + \{x\}) \subseteq \mathbf{A}(S + T)$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$S_1 \subset S_2$ 时，在 S_1 中见证某渐近方向的点列也在 S_2 中，所以 $AS_1 \subset AS_2$。若 $T \neq \varnothing$，取固定 $a \in T$，则 $S + a \subset S + T$；结合平移不变性，$AS = A(S + a) \subset A(S + T)$。这里 a 必须固定，不能随序列任意变化。乘积的结论一般只是包含，原因是各分量需要共同的缩放尺度，见第33页。

31 原课件第 31 页：平移不改变渐近锥：逐步证明

Vectors in $\mathbb{R}^N$

(5) If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S+\{x\}) = \mathbf{A}S$.

Proof. We just need to prove $\mathbf{A}(S+\{x\}) \subseteq \mathbf{A}S$, then to prove $\mathbf{A}S \subseteq \mathbf{A}(S+\{x\})$, we just need to let $S = (S+\{x\}) + \{-x\}$ and we arrive at the conclusion. Let $T = S + \{x\}$.

For any $d \in \mathbf{A}(S+\{x\})$, $d \in (S+\{x\})_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq S + \{x\}$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. Let $\{x'_k = x_k - x\} \subseteq S$ and

$$\left\| \frac{x'_k}{t_k} - d \right\| \leqq \left\| \frac{x'_k}{t_k} - \frac{x_k}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| = \left\| \frac{x}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| \to 0$$

hence $\lim\limits_{k\to\infty} \dfrac{x'_k}{t_k} = d$, i.e., $d \in S_\infty$, hence $d \in \mathbf{A}S$. Hence $\mathbf{A}(S+\{x\}) \subseteq \mathbf{A}S$. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 $d \in (S+a)_\infty$，存在 $z_k \in S + a$、$t_k \to \infty$ 使 $z_k/t_k \to d$。写 $z_k = x_k + a$，$x_k \in S$，则

$$\|x_k/t_k - d\| \leq \|z_k/t_k - d\| + \|a\|/t_k \to 0.$$

故 $(S+a)_\infty \subset S_\infty$。把 $S+a$ 当作新集合，再平移 $-a$，得到反向包含。直观是固定距离的搬动除以越来越大的尺度后消失。

32 原课件第 32 页：平移不变性的等价链写法

Vectors in $\mathbb{R}^N$

(5) If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S + \{x\}) = \mathbf{A}S$.

Proof.

$$
\begin{aligned}
& d \in \mathbf{A}(S) \Longleftrightarrow d \in S_\infty \\
\Longleftrightarrow \quad & \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq S \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \\
\Longleftrightarrow \quad & \text{there exist } \{x_k + x; k \in \mathbb{N}\} \subseteq S + \{x\} \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k + x}{t_k} = d \\
\Longleftrightarrow \quad & d \in (S + \{x\})_\infty \Longleftrightarrow d \in \mathbf{A}(S + \{x\})
\end{aligned}
$$

□

In other words, a translation of S does not alter its asymptotic cone.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页将第31页的双包含压缩为等价关系。核心等式是 $(x_k + a)/t_k = x_k/t_k + a/t_k$，最后一项趋于 0，所以两边具有同一个极限。练习时先自行写出 x_k 属于哪个集合，再写极限；不必背每一行排版。若平移量变为 a_k 且 a_k/t_k 不趋零，结论便可能改变。

33 原课件第 33 页：乘积的渐近锥包含于各渐近锥的乘积

Vectors in $\mathbb{R}^N$

(7) If, for every $k = 1, \cdots, K$, S_k is a subset of $\mathbb{R}^{N_k}$, then $\mathbf{A}\left(\prod_{k=1}^{K} S_k\right) \subseteq \prod_{k=1}^{K} \mathbf{A}S_k$.

Proof. For $d \in \mathbf{A}\left(\prod_{n=1}^{N} S_n\right)$. If $d = 0$, then $d \in \prod_{n=1}^{N} \mathbf{A}S_n$ is trivial. If $d \neq 0$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq \prod_{n=1}^{N} S_n$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. Let

$$x_k = \left(x_k^1, \cdots, x_k^n, \cdots, x_k^N\right), d = \left(d^1, \cdots, d^n, \cdots, d^N\right)$$

where $x_k^n \in S_n$ and $d^n \in \mathbb{R}$ for $n = 1, \cdots, N$. For each $n = 1, \cdots, N$, we have $\lim_{k\to\infty} \frac{x_k^n}{t_k} = d^n$. Hence $d^n \in \mathbf{A}S_n$, hence $d \in \prod_{n=1}^{N} \mathbf{A}S_n$. Thus, $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 $C = S_1 \times \cdots \times S_K$ 非空，$d \in C_\infty$。取 $x_q = (x_{1,q}, \ldots, x_{K,q}) \in C$、$t_q \to \infty$ 使 $x_q/t_q \to d = (d_1, \ldots, d_K)$。向量收敛蕴含每个坐标块收敛，所以 $x_{k,q}/t_q \to d_k$，即 $d_k \in (S_k)_\infty$。因此

$$C_\infty \subset \prod_{k=1}^{K} (S_k)_\infty.$$

反向不自动成立：各个方向的见证序列可能只在互不匹配的尺度出现。对凸集，可由第105–108页的渐近正则性解决尺度匹配问题。

34 原课件第 34 页：两个核心应用：交有界，和闭



Vectors in $\mathbb{R}^N$

p. It is now possible to give answers to the two important questions: When is the intersection of a collection of sets bounded? When is the sum of a finite number of closed sets closed? One can prove the intuitive result:

(8) Given a collection of subsets of $\mathbb{R}^N$, if the intersection of their asymptotic cones is $\{0\}$, then their intersection is bounded.

One can also prove:

(9) Given K closed subsets of $\mathbb{R}^N$, if their asymptotic cones are positively semi-independent, then their sum is closed.

This is a generalization of (2), since the asymptotic cone of a bounded set in $\{0\}$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

交集有界的充分条件是所有渐近锥的交仅为 $\{0\}$；闭集之和闭的充分条件是它们的渐近锥正半独立。两者都在阻止某种“逃到无穷”的行为：前者阻止同时在所有集合中逃走，后者阻止各分量逃走但总和相互抵消。它们是充分条件，不要写成必要条件。例如整个 $\mathbb{R}$ 加自己仍闭，两个渐近锥却能抵消。

35 原课件第 35 页：交集有界：用单调性证明

Vectors in $\mathbb{R}^N$

(8) Given a collection of subsets of $\mathbb{R}^N$, if the intersection of their asymptotic cones is $\{0\}$, then their intersection is bounded.

Proof. As we have that "$S_1 \subseteq S_2$" implies "$\mathbf{A}S_1 \subseteq \mathbf{A}S_2$", then, for an arbitrary index set Γ, let $C_\gamma \subseteq \mathbb{R}^N$, $\gamma \in \Gamma$, we have

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \mathbf{A}C_\gamma$$

for all $\gamma \in \Gamma$. Hence

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma$$

If $\bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma = \{0\}$, then $\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) = \{0\}$. By Proposition 2.3, we have $\bigcap_{\gamma\in\Gamma} C_\gamma$ is bounded. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

令 $D = \bigcap_{\gamma\in I} C_\gamma$。若 $D = \varnothing$，有界性直接成立。若非空，则 $D \subset C_\gamma$ 给 $D_\infty \subset (C_\gamma)_\infty$ 对每个 γ 成立，因此

$$D_\infty \subset \bigcap_{\gamma} (C_\gamma)_\infty = \{0\}.$$

因为 D 非空，$0 \in D_\infty$，于是 $D_\infty = \{0\}$，第29页即给 D 有界。这就是“先处理空集，再使用非空定理”的标准写法。

36 原课件第 36 页：不能从递减交直接跳到某个截尾为空

Vectors in $\mathbb{R}^N$

(8) For all $\gamma \in \Gamma$, $S_\gamma \subseteq \mathbb{R}^N$, if $\bigcap_{\gamma\in\Gamma} \mathbf{A}(S_\gamma) = \{0\}$, then $\bigcap_{\gamma\in\Gamma} S_\gamma$ is bounded.

Proof. For all $\gamma \in \Gamma$, $\bigcap_{\gamma\in\Gamma} S_\gamma \subseteq S_\gamma$, then $\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right) \subseteq \mathbf{A}(S_\gamma)$.

Whence $\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right) \subseteq \bigcap_{\gamma\in\Gamma} \mathbf{A}(S_\gamma) = \{0\}$. There exists $M_0 \geqq 0$ such that $\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)^M = \varnothing$ for $M \geqq M_0$, thus $\bigcap_{\gamma\in\Gamma} S_\gamma$ is bounded.

□

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页用截尾集合重述第35页。结论正确，但中间“存在 M_0 使截尾为空”需要第29页的有界性判别，不能只由递减集合交为空或交为 $\{0\}$ 直接断言。一般递减非空集可以有空交，例如 $[n,\infty)$。在这里，可反证：若交集无界，就抽出单位极限方向，同时属于全部渐近锥，与仅含零矛盾。

37 原课件第 37 页：线性映射何时把闭集送到闭集



Theorem 3.5. (Images under Linear Mappings) For a linear mapping $L : \mathbb{R}^N \to \mathbb{R}^K$ and a closed set $C \subseteq \mathbb{R}^N$, a sufficient condition for the set $L(C)$ to be closed is that $L^{-1}\{0\} \cap C_\infty = \{0\}$. Under this condition one has $L(C_\infty) = L(C)_\infty$, although in general only $L(C_\infty) \subseteq L(C)_\infty$.

Theorem 3.5. (Images under Linear Mappings) For a linear mapping $L : \mathbb{R}^N \to \mathbb{R}^K$ and a closed set $C \subseteq \mathbb{R}^N$, if $L^{-1}\{0\} \cap C_\infty = \{0\}$, then

(1) the set $L(C)$ is closed;

(2) $L(C_\infty) = L(C)_\infty$.

Although in general only $L(C_\infty) \subseteq L(C)_\infty$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

线性映射 L 满足 $L(\alpha x+\beta y)=\alpha L(x)+\beta L(y)$；有限维时它连续。核 $\ker L = L^{-1}(\{0\})$ 是所有被压到零的方向。条件

$$\ker L \cap C_\infty = \{0\}$$

表示 C 的非零逃逸方向不能被 L 完全压扁。若 C 闭，则它保证 $L(C)$ 闭，而且 $L(C_\infty)=(L(C))_\infty$。无条件成立的方向只有 $L(C_\infty)\subset(L(C))_\infty$。下面四页分别处理“防逃逸、闭性、正向包含、反向包含”。

38 原课件第 38 页：线性像证明的核心：归一化反证

Vectors in $\mathbb{R}^N$

Proof. The condition will be shown first to imply that L carries unbounded sequences in C into unbounded sequences in $\mathbb{R}^K$. If this were not true, there would exist a sequence of points $\{x_k; k \in \mathbb{N}\} \subseteq C$ such that $\|x_k\| \to \infty$, but such that the sequence of images $y_k = L(x_k)$ is bounded. As $\left\{\frac{x_k}{\|x_k\|}; k \in \mathbb{N}\right\}$ is bounded, so we can extract a subsequence of $\left\{\frac{x_k}{\|x_k\|}; k \in \mathbb{K}\right\}$ such that $\lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{x_k}{\|x_k\|} = x$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|x\| = 1$. Hence $x \in C_\infty$, but also (because linear mappings are continuous)

$$L(x) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{1}{\|x_k\|} L(x_k) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{y_k}{\|x_k\|} = 0$$

which the condition prohibits.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若存在 $x_k \in C$，$\|x_k\| \to \infty$，而 $L(x_k)$ 有界，则单位向量取子列 $x_k/\|x_k\| \to u$，$\|u\| = 1$。按定义 $u \in C_\infty$；连续性和线性给

$$L(u) = \lim_k \frac{L(x_k)}{\|x_k\|} = 0.$$

于是 $u \in \ker L \cap C_\infty$ 且非零，与条件矛盾。更准确的可用结论是：C 中任何像有界的点列自身必有界；若自身无界，先抽取范数趋无穷的子列，再运用本证明。

39 原课件第 39 页：线性像闭：把收敛提升回原集合



Assuming henceforth that the condition holds, we prove now that $L(C)$ is a closed set. Suppose $y_k \to y$ with $y_k \in L(C)$, or in other words, $y_k = L(x_k)$ for some $x_k \in C$. The sequence $\{L(x_k)\}$ is bounded, so the sequence $\{x_k; k \in \mathbb{N}\}$ must be bounded as well, for the reasons just given, and it therefore has a cluster point x. We have $x \in C$ (because C is closed) and $y = L(x)$ (again because linear mappings are continuous), and therefore $y \in L(C)$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

任取 $y_k \in L(C)$ 且 $y_k \to y$，每次选一个原像 $x_k \in C$ 满足 $L(x_k) = y_k$。(y_k) 有界，第38页使 (x_k) 有界；抽取 $x_{k_j} \to x$。C 闭给 $x \in C$，连续性给 $L(x) = \lim_j y_{k_j} = y$，故 $y \in L(C)$。注意 L 可能不单射，原像无须唯一；任意选取的原像都受这里的有界性论证控制。

40 原课件第 40 页：渐近方向先取像：总是可行

Vectors in $\mathbb{R}^N$

The next task is to show that $L(C_\infty) \subseteq L(C)_\infty$. This is trivial when $C = \varnothing$, so we assume $C \neq \varnothing$. Any $d \in C_\infty$, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. The points $y_k = L(x_k)$ in $L(C)$ and

$$\lim_{k\to\infty} \frac{y_k}{t_k} = \lim_{k\to\infty} \frac{L(x_k)}{t_k} = L\left(\lim_{k\to\infty} \frac{x_k}{t_k}\right) = L(d)$$

hence $L(d) \in L(C)_\infty$. Hence $L(C_\infty) \subseteq L(C)_\infty$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

任取 $d \in C_\infty$，选 $x_k \in C$、$t_k \to \infty$ 使 $x_k/t_k \to d$。由于 $L(x_k) \in L(C)$ 且

$$\frac{L(x_k)}{t_k} = L\left(\frac{x_k}{t_k}\right) \longrightarrow L(d),$$

得 $L(d) \in (L(C))_\infty$。这一步既不使用 C 闭，也不使用核条件。考试中指出哪些假设用在哪里，能避免把强条件当成所有步骤的理由。

41 原课件第 41 页：线性像渐近锥的反向包含

Vectors in $\mathbb{R}^N$

For the inclusion $L(C_\infty) \supseteq L(C)_\infty$, we consider any $y \in L(C)_\infty$, there exist $\{y_k; k \in \mathbb{N}\} \subseteq L(C)$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{y_k}{t_k} = y$. For any choice of $x_k \in C$ with $L(x_k) = y_k$, we have $\lim_{k\to\infty} L\left(\frac{x_k}{t_k}\right) = y$. The boundedness of the sequence $\left\{L\left(\frac{x_k}{t_k}\right); k \in \mathbb{N}\right\}$ implies the boundedness of the sequence $\left\{\frac{x_k}{t_k}; k \in \mathbb{N}\right\}$, which therefore must have a cluster point x. Then $x \in C_\infty$ and $L(x) = y$, and we conclude that $y \in L(C_\infty)$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

须补严谨步骤，不能直接把第38页用于不在 C 内的 x_n/t_n。若 $Lx_n/t_n \to y$，假设 x_n/t_n 无界，则抽子列使 $r_n = \|x_n\|/t_n \to \infty$。归一化后再抽 $x_n/\|x_n\| \to d$，其中 $\|d\| = 1$，$d \in C_\infty$ 且 $Ld = \lim(Lx_n/t_n)/r_n = 0$，违反核条件。因此 x_n/t_n 有界，抽子列趋于 d，定义给 $d \in C_\infty$ 且 $Ld = y$。

42 原课件第 42 页：闭集和与正半独立

Vectors in $\mathbb{R}^N$

Theorem 3.6 (Sums of Sets). For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, a sufficient condition for $\sum_{n=1}^{N} C_n$ to be closed is the non-existence of vectors $x_n \in (C_n)_\infty$ such that $\sum_{n=1}^{N} x_n = 0$, except with $x_n = 0$ for all n. Then also

$$\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty .$$

Theorem 3.6 (Sums of Sets). For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, if $(C_1)_\infty, \cdots, (C_N)_\infty$ are positively semi-independent (P.S.I.), then

(1) $\sum_{n=1}^{N} C_n$ is closed;

(2) $\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

正半独立指若 $d_i \in (C_i)_\infty$ 且 $\sum_i d_i = 0$，则每个 $d_i = 0$。设积集 $C = \prod_i C_i$，线性映射 $L(x_1, \ldots, x_m) = \sum_i x_i$。逐坐标极限给 $C_\infty \subseteq \prod_i (C_i)_\infty$，所以核条件成立。前面的线性像定理给 $L(C) = \sum_i C_i$ 闭，且 $(\sum_i C_i)_\infty = L(C_\infty) \subseteq \sum_i (C_i)_\infty$。这说明渐近方向不能互相抵消是保证闭和的重要条件；闭集的和一般不自动闭。

43 原课件第 43 页：闭集和与正半独立

Vectors in $\mathbb{R}^N$

Proof. Let $C = \prod_{n=1}^{N} C_n$. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n$$

for $x_n \in C_n$. Then L is a linear mapping. As Theorem 3.3 shows, $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$. And by the condition, we have

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}$$

hence

$$L^{-1}(0) \cap C_\infty = \{0\}$$

so by Theorem 3.5, we have $L(C)$ is closed. As $L(C) = \sum_{n=1}^{N} C_n$, we have that $\sum_{n=1}^{N} C_n$ is closed.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭集和与正半独立”。完整定义、条件和论证见本讲原页第 42 页下方。

44 原课件第 44 页：闭集和与正半独立

Vectors in $\mathbb{R}^N$

And again by Theorem 3.5, $L(C_\infty) = L(C)_\infty$, i.e.

$$L\left(\left(\prod_{n=1}^{N} C_n\right)_\infty\right) = \left(\sum_{n=1}^{N} C_n\right)_\infty$$

We also have

$$L\left(\prod_{n=1}^{N} (C_n)_\infty\right) = \sum_{n=1}^{N} (C_n)_\infty$$

and

$$\left(\prod_{n=1}^{N} C_n\right)_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$$

hence

$$\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty$$

□

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭集和与正半独立”。完整定义、条件和论证见本讲原页第 42 页下方。

45 原课件第 45 页：闭集和与正半独立

Vectors in $\mathbb{R}^N$

(9) Given K closed subsets of $\mathbb{R}^N$, if their asymptotic cones are positively semi-independent, then their sum is closed.

Proof. Let $C = \prod_{n=1}^{N} C_n$. Then $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$ from Theorem 3.3. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n, \qquad \text{for} \quad x_n \in C_n,$$

then L is a linear mapping.

If $(C_1)_\infty, \cdots, (C_N)_\infty$ are positively semi-independent (P.S.I.), then

$$x_n \in (C_n)_\infty \text{ for } n = 1, \cdots, N \quad \text{and} \quad \sum_{n=1}^{N} x_n = 0 \quad \text{implies} \quad x_n = 0 \text{ for } n = 1, \cdots, N$$

$$(x_1, \cdots, x_N) \in \prod_{n=1}^{N} (C_n)_\infty \quad \text{and} \quad L((x_1, \cdots, x_N)) = 0 \quad \text{implies} \quad (x_1, \cdots, x_N) = \mathbf{0}$$

$$(x_1, \cdots, x_N) \in L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty \quad \text{implies} \quad (x_1, \cdots, x_N) = \mathbf{0}$$

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}$$

hence

$$L^{-1}(0) \cap C_\infty = \{0\}.$$

For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, C is closed. By Theorem 3.5, we have $L(C) = \sum_{n=1}^{N} C_n$ is closed.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭集和与正半独立”。完整定义、条件和论证见本讲原页第 42 页下方。

46 原课件第 46 页：凸集与多面锥



q. A subset of $\mathbb{R}^N$ is convex if it contains the closed segment $[x_1, x_2]$ whenever it contains the points x_1, x_2; in other words, if, whenever it contains the points x_1, x_2, it contains their weighted average with arbitrary positive weights. The set K in fig. 7.a, 7.b, o, 7.c is convex. The graph of fig. 2.a or 2 .b is not convex. The condition on a set that it be convex is crucial for economics.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

凸指任取 $x, y \in C$ 及 $0 \le t \le 1$，$tx + (1-t)y \in C$。这是整条线段都在集合中；不是只检验中点。多面锥由有限向量的非负线性组合构成，权重不要求和为1；凸组合则要求和为1。二者必须区分。

47 原课件第 47 页：凸集与多面锥

Vectors in $\mathbb{R}^N$

r. A convex polyhedral cone is the sum of K closed half-lines.

Figure 6 pictures a convex polyhedral cone in $\mathbb{R}^2$ which is the sum of four, closed half-lines with origin 0. L_1 and L_3 are imagined to be in the plane of the page, L_2 in front of it, L_4 behind it.

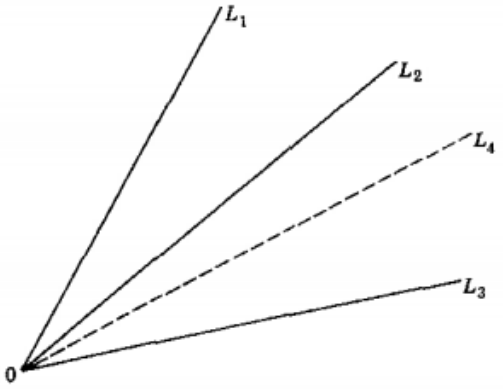


Fig. 6

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“凸集与多面锥”。完整定义、条件和论证见本讲原页第 46 页下方。

48 原课件第 48 页：凸性的基本运算及连通性



Vectors in $\mathbb{R}^N$

s. It is easy to prove that:

(10) the intersection of a set of convex sets is convex,

(11) the sum and the product of K convex sets are convex,

(12) the adherence of a convex set is convex,

and also, using (8) of 1.6.u, that:

(13) a convex set is connected.

One can prove that:

(14) a closed, convex set owning 0, contains its asymptotic cone.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

交：两端点属于每个凸因子，线段逐个属于它们，故属于交。和：写 $x=\sum x_i, y=\sum y_i$，则 $tx+(1-t)y=\sum(tx_i+(1-t)y_i)$ 属于和；积逐坐标同理。闭包：选 $x_n, y_n \in C$ 趋于闭包两点，凸组合在 C，极限在闭包。凸集连通：若被相对开集分离，连接两侧点的连续路径 $t \mapsto tx+(1-t)y$ 把分离拉回 $[0,1]$，与区间连通矛盾。最后若 C 闭凸且含0，$x_n/t_n \to d$ 且 $t_n \to \infty$，充分大时 $x_n/t_n=(1-1/t_n)0+x_n/t_n \in C$，闭性给 $d \in C$。

49 原课件第 49 页：凸性的基本运算及连通性



(10) the intersection of a set of convex sets is convex.

Proof of (10). If $\{X_\gamma; \gamma \in \Gamma\}$ is a family of convex subsets of X. Then for any x_1 and x_2 in $\bigcap_{\gamma\in\Gamma} X_\gamma$, then for any $\gamma \in \Gamma$, we have x_1 and x_2 in X_γ. For any $0 \leqslant t \leqslant 1$, we have $tx_1 + (1-t)x_2 \in X_\gamma$, hence $tx_1 + (1-t)x_2 \in \bigcap_{\gamma\in\Gamma} X_\gamma$. So $\bigcap_{\gamma\in\Gamma} X_\gamma$ is convex. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“凸性的基本运算及连通性”。完整定义、条件和论证见本讲原页第 48 页下方。

50 原课件第 50 页：凸性的基本运算及连通性

Vectors in $\mathbb{R}^N$

(11) the sum and the product of K convex sets are convex.

Proof of (11). Let $X_1, \cdots, X_K$ be convex. For any x^1 and x^2 in $\sum_{k=1}^{K} X_k$, there exist $x_k^1 \in X_k$ and $x_k^2 \in X_k$ for $k = 1, \cdots, K$ such that $x^1 = \sum_{k=1}^{K} x_k^1$ and $x^2 = \sum_{k=1}^{K} x_k^2$. For any $t \in [0,1]$, we have

$$tx^1 + (1-t)\,x^2 = t\sum_{k=1}^{K} x_k^1 + (1-t)\sum_{k=1}^{K} x_k^2 = \sum_{k=1}^{K}\left[tx_k^1 + (1-t)\,x_k^2\right] \in \sum_{k=1}^{K} X_k$$

hence $\sum_{k=1}^{K} X_k$ is convex.

Let $x^i \in \prod_{k=1}^{K} X_k$ with $x^i = \left(x_1^i, \cdots, x_k^i, \cdots, x_K^i\right)$, $i =, 2$. For any $t \in [0,1]$, we have

$$\begin{aligned} & tx^1 + (1-t)\,x^2 = t\left(x_1^1, \cdots, x_k^1, \cdots, x_K^1\right) + (1-t)\left(x_1^2, \cdots, x_k^2, \cdots, x_K^2\right) \\ = \quad & \left(tx_1^1 + (1-t)x_1^2, \cdots, tx_k^1 + (1-t)x_k^2, \cdots, tx_K^1 + (1-t)x_K^2\right) \in \prod_{k=1}^{K} X_k \end{aligned}$$

hence $\prod_{k=1}^{K} X_k$ is convex. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“凸性的基本运算及连通性”。完整定义、条件和论证见本讲原页第 48 页下方。

51 原课件第 51 页：凸性的基本运算及连通性

Vectors in $\mathbb{R}^N$

(12) the adherence of a convex set is convex.

Proof of (12). Let x^1 and x^2 in convex set $\overline{X}$. By the definition of closure, there exist sequences $\left\{x_q^1\right\} \subseteq X$ and $\left\{x_q^2\right\} \subseteq X$ with limits x^1 and x^2, respectively. Now, because X is convex, we know that for each x_q^1 and x_q^2, and for all $t \in [0,1]$, we have $tx_q^1 + (1-t)x_q^2 \in X$. This establishes the sequence $\left\{tx_q^1 + (1-t)x_q^2\right\} \subseteq X$ which is convergent by the convergence of $\left\{x_q^1\right\}$ and $\left\{x_q^2\right\}$. So we have that $tx^1 + (1-t)\,x^2 \in \overline{X}$, as $\overline{X}$ includes all limit points of X. This establishes that $\overline{X}$ is convex. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“凸性的基本运算及连通性”。完整定义、条件和论证见本讲原页第 48 页下方。

52 原课件第 52 页：凸性的基本运算及连通性

Vectors in $\mathbb{R}^N$

(13) a convex set is connected.

Proof of (13). Suppose $E \subseteq \mathbb{R}^N$ is convex but not connected, then there exist non-empty open sets U_1 and U_2 in E such that $U_1 \cup U_2 = E$ and $U_1 \cap U_2 = \varnothing$. Let $x_1 \in U_1$ and $x_2 \in U_2$, We define

$$V_1 = \left\{ t \in [0,1] : tx^1 + (1-t)\, x^2 \in U_1 \right\}$$

and

$$V_2 = \left\{ t \in [0,1] : tx^1 + (1-t)\, x^2 \in U_2 \right\}$$

then $V_1 \neq \varnothing$ and $V_2 \neq \varnothing$ and $V_1 \cup V_2 = [0,1]$ and $V_1 \cap V_2 = \varnothing$. As U_1 and U_2 are open, hence V_1 and V_2 are also open on $[0,1]$. Hence $[0,1]$ is disconnected, contradicting (8) of 1.6.u. Hence E is connected. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“凸性的基本运算及连通性”。完整定义、条件和论证见本讲原页第 48 页下方。

53 原课件第 53 页：凸性的基本运算及连通性

Vectors in $\mathbb{R}^N$

(14) a closed, convex set owning 0, contains its asymptotic cone.

Proof of (14). Let $y \in \mathbf{A}C$. For any $M \geqq 0$, $\mathbb{R}_+ y \in \Gamma\left(C^M\right)$.

(1) $y \in C^M$, then $y \in C$.

(2) $y \notin C^M$, then there exists $\lambda < 1$ such that $\frac{1}{\lambda} y \in C^M \subseteq C$,

hence $y = (1-\lambda)0 + \lambda \frac{1}{\lambda} y \in C$.

Therefore $y \in \mathbf{A}C$ implies $y \in C$. That is, $\mathbf{A}C \subseteq C$. □

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本页继续“凸性的基本运算及连通性”。完整定义、条件和论证见本讲原页第 48 页下方。

54 原课件第 54 页：凸性的基本运算及连通性

Vectors in $\mathbb{R}^N$

(14) a closed, convex set owning 0, contains its asymptotic cone.

Proof of (14). Assume $d \in C_\infty$: By definition of the asymptotic cone, there exists a sequence $\{x_k\} \subseteq C$ and a sequence $\{t_k\} \subseteq \mathbb{R}$ with $t_k \to \infty$, such that $\dfrac{x_k}{t_k} \to d$.

- Since C is convex and contains 0, for each k, the point $\lambda_k x_k$ lies in C for $\lambda_k \in [0,1]$. Choose $\lambda_k = \dfrac{1}{t_k}$, then $\lambda_k x_k = \dfrac{x_k}{t_k}$.
- As $t_k \to \infty$, $\lambda_k \to 0$, and $\dfrac{x_k}{t_k} \to d$.
- Since C is closed and $\dfrac{x_k}{t_k} \in C$ for all k, the limit d must also belong to C, $d \in C$.

Since $d \in C_\infty$ implies $d \in C$, we have $C_\infty \subseteq C$.

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55 原课件第 55 页：凸包与和的交换及证明纠错

Vectors in $\mathbb{R}^N$

t. Let S he a subset of $\mathbb{R}^N$. Its convex hull, denoted $\dot{S}$, is defined as the intersection of all the convex sets containing S. According to (10), $\dot{S}$ is indeed convex and can therefore also be characterized as the smallest convex set containing S. One can prove that:

(15) Given K subsets of $\mathbb{R}^N$, the convex hull of their sum is equal to the sum of their convex hulls.

The closed convex hull of S is, by definition, the adherence $\overline{\dot{S}}$ of the convex hull of S. According to (3) of 1.6.j, $\overline{\dot{S}}$ is indeed closed; according to (12), it is indeed convex. It is easy to see that a closed convex set containing S necessarily contains $\overline{\dot{S}}$, which can therefore also he characterized as the smallest closed convex set containing S.

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凸包是所有包含 S 的凸集的交，等价于有限凸组合全体。闭凸包是它的闭包。证明 $\operatorname{co}(A+B)=\operatorname{co}A+\operatorname{co}B$：一边把 $\sum\lambda_i(a_i+b_i)$ 分成两凸组合；另一边若 $a=\sum_i\alpha_i a_i, b=\sum_j\beta_j b_j$，则 $a+b=\sum_{i,j}\alpha_i\beta_j(a_i+b_j)$，积权重非负且总和1。有限多集合用权重 $\prod_k\lambda_{ki_k}$。第58页直接把总权重 K 归一化会把目标点也除以 K，因此不成立；第59页积权重论证才正确。

56 原课件第 56 页：凸包与和的交换及证明纠错

Vectors in $\mathbb{R}^N$

(15) Given K subsets of $\mathbb{R}^N$, the convex hull of their sum is equal to the sum of their convex hulls.

(15) If $S_k \subseteq \mathbb{R}^N$ for $k = 1, \cdots, K$, denote $S = \sum_{k=1}^{K} S_k$, then $\dot{S} = \sum_{k=1}^{K} \dot{S}_k$.

$$\operatorname{conv}(A) = \left\{ \sum_{i=1}^{I} \lambda_i x_i \mid x_i \in A, \lambda_i \geqq 0, \sum_{i=1}^{I} \lambda_i = 1, I \in \mathbb{N} \right\}.$$

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本页继续“凸包与和的交换及证明纠错”。完整定义、条件和论证见本讲原页第 55 页下方。

57 原课件第 57 页：凸包与和的交换及证明纠错

Vectors in $\mathbb{R}^N$

Proof of (15). Prove $\dot{S} \subseteq \sum_{k=1}^{K} \dot{S}_k$.

Let $x \in \dot{S}$. By definition, x is a convex combination of points in $S = \sum_{k=1}^{K} S_k$. $x = \sum_{i=1}^{I} \lambda_i \sum_{k=1}^{K} x_{ki}$, where $x_{ki} \in S_k$, $\lambda_i \geqq 0$, and $\sum_{i=1}^{I} \lambda_i = 1$. Rearranging the sum, we have $x = \sum_{k=1}^{K} \sum_{i=1}^{I} \lambda_i x_{ki}$. Each term $\sum_{i=1}^{I} \lambda_i x_{ki}$ is a convex combination of points in S_k, so $\sum_{i=1}^{I} \lambda_i x_{ki} \in \dot{S}_k$. Therefore $x \in \sum_{k=1}^{K} \dot{S}_k$. This proves $\dot{S} \subseteq \sum_{k=1}^{K} \dot{S}_k$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“凸包与和的交换及证明纠错”。完整定义、条件和论证见本讲原页第 55 页下方。

58 原课件第 58 页：凸包与和的交换及证明纠错

Vectors in $\mathbb{R}^N$

Proof of (15). Prove $\sum_{k=1}^{K} \dot{S}_k \subseteq \dot{S}$.

Let $x \in \sum_{k=1}^{K} \dot{S}_k$. By definition, $x = \sum_{k=1}^{K} x_k$, where $x_k \in \dot{S}_k$ for $k = 1, \cdots, K$. Each x_k is a convex combination of points in S_k. $x_k = \sum_{i_k=1}^{I_k} \lambda_{ki_k} x_{ki_k}$, where $x_{ki_k} \in S_k$, $\lambda_{ki_k} \geqq 0$, and $\sum_{i_k=1}^{I_k} \lambda_{ki_k} = 1$.

Substituting into x, we have $x = \sum_{k=1}^{K} \sum_{i_k=1}^{I_k} \lambda_{ki_k} x_{ki_k}$. This is a convex combination of points in $S = \sum_{k=1}^{K} S_k$, because $\sum_{k=1}^{K} \sum_{i=1}^{I_k} \lambda_{ki} = \sum_{k=1}^{K} 1 = K$ and we can normalize the coefficients to form a convex combination. Therefore $x \in \dot{S}$. This proves $\sum_{k=1}^{K} \dot{S}_k \subseteq \dot{S}$.

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本页继续“凸包与和的交换及证明纠错”。完整定义、条件和论证见本讲原页第 55 页下方。

59 原课件第 59 页：凸包与和的交换及证明纠错

Vectors in $\mathbb{R}^N$

$$
\begin{aligned}
x &= \sum_{k=1}^{K}\sum_{i_k=1}^{I_k}\lambda_{ki_k}x_{ki_k} = \sum_{i_1=1}^{I_1}\lambda_{1i_1}x_{1i_1} + \cdots + \sum_{i_K=1}^{I_K}\lambda_{Ki_K}x_{Ki_K} \\
&= \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}x_{1i_1} + \cdots + \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}x_{Ki_K} \\
&= \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}\left(x_{1i_1} + \cdots + x_{Ki_K}\right) = \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}\sum_{k=1}^{K}x_{ki_k} \\
&= \sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\left[\lambda_{1i_1}\cdots\lambda_{Ki_K}\right]\sum_{k=1}^{K}x_{ki_k} = \sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\left[\prod_{k=1}^{K}\lambda_{ki_k}\right]\left[\sum_{k=1}^{K}x_{ki_k}\right]
\end{aligned}
$$

where $\sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\prod_{k=1}^{K}\lambda_{ki_k} = \sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{1i_1}\cdots\lambda_{Ki_K} = \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K} = 1$, then $x \in \dot{S}$. □

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本页继续“凸包与和的交换及证明纠错”。完整定义、条件和论证见本讲原页第 55 页下方。

60 原课件第 60 页：内积、超平面与半空间

Vectors in $\mathbb{R}^N$

u. Let $x^1 = \left(x_n^1\right)$ and $x^2 = \left(x_n^2\right)$ be two elements of $\mathbb{R}^N$; one defines their inner product $x^1 \cdot x^2$ as the number $\sum_{n=1}^{N} x_n^1 x_n^2$. It is clear that the function $\left(x^1, x^2\right) \to x^1 \cdot x^2$ from $\mathbb{R}^N \times \mathbb{R}^N$ to $\mathbb{R}$ is continuous.

When $x^1 \cdot x^2 = 0$. one says that x^1 and x^2 are orthogonal. In the case of $\mathbb{R}^2$, visualized as in 1.6.c. saying that two vectors x^1 and x^2 different from 0 are orthogonal is equivalent to saying that the half-lines $0, x^1$ and $0, x^2$ are perpendicular, provided that the units on the three axes have the same length.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$p \cdot x = \sum_i p_i x_i$ 为内积。固定非零 p，$H = \{x : p \cdot x = c\}$ 为超平面，p 为法向量；非零条件避免整个空间或空集。两闭半空间由 $\geq c$ 与 $\leq c$ 定义。线性函数连续，弱不等式在极限下保持，故闭；线性组合保持同侧不等式，故凸。

61 原课件第 61 页：内积、超平面与半空间



v. Let p he an element of $\mathbb{R}^N$ different from 0, and c a real number; the set $H = \left\{z \in \mathbb{R}^N \mid p \cdot z = c\right\}$ is a hyperplane with normal p. If z' is a point in H (one says also that H goes through z'), one has $p \cdot z' = c$ and the above expression may be rewritten $H = \left\{z \in \mathbb{R}^N \mid p \cdot (z - z') = 0\right\}$. Thus the hyperplane H is the set of points z of $\mathbb{R}^N$ such that $z - z'$ is orthogonal to p; p and H are said to be orthogonal. If p and c are multiplied by the same real number different from 0, the hyperplane H is unchanged. An intersection of hyperplanes is called a linear manifold.

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本页继续“内积、超平面与半空间”。完整定义、条件和论证见本讲原页第 60 页下方。

62 原课件第 62 页：内积、超平面与半空间



w. Given a hyperplane H with normal p, the point z of $\mathbb{R}^N$ is said to be above H if $p \cdot z > c$. The closed half-space above H is $\{z \in \mathbb{R}^N \mid p \cdot z \geqq c\}$. One obtains similar definitions replacing above, $>, \geqq$ by below, $<, \leqq$. A closed half-space is easily seen to be closed and convex. So is a hyperplane, since it is the intersection of two closed half-spaces, and hence a linear manifold.

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本页继续“内积、超平面与半空间”。完整定义、条件和论证见本讲原页第 60 页下方。

63 原课件第 63 页：支撑超平面的完整证明



Vectors in $\mathbb{R}^N$

x. A hyperplane H is said to be bounding for a subset S of $\mathbb{R}^N$ if S is contained in one of the two closed half-spaces determined by H. In other words, H is bounding for S if S is entirely on one side of H with, possibly, points in it. One can prove the fundamental theorem:

(16) (Minkowski's Theorem) Let K be a convex subset of $\mathbb{R}^N$ and x a point of $\mathbb{R}^N$. There is a hyperplane H through x and bounding for K if and only if x is not interior to K.

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要证凸集 K 在非内点 x 有过 x 的界超平面。先对非空闭凸 C 与外点 z 求最近点 a：最小化距离可限制到一个闭有界球，与 C 的交紧，故最小存在。对任意 $y\in C$，线段 $a+t(y-a)$ 可行，比较平方距离并除以 $t>0$ 再令 $t\downarrow 0$，得 $(z-a)\cdot(y-a)\le 0$。若 $x\notin C$，取 $p=x-a$，则 $p\cdot(y-x)\le -\|p\|^2<0$，过 x 的面已界住 C。若 $x\in\partial C$，取外点 $z_n\to x$ 及最近点 a_n，有 $\|a_n-x\|\le 2\|z_n-x\|\to 0$。单位法向 $p_n=(z_n-a_n)/\|z_n-a_n\|$ 抽子列趋于单位 p，对每个 $y\in C$ 取极限得 $p\cdot(y-x)\le 0$。一般凸 K 令 $C=\overline{K}$。若内域为空，则 K 的仿射包为真子空间（否则有 $N+1$ 个仿射独立点，其单纯形有内域），取与其方向正交的非零法向即可过任意 x 界住 K。若内域非空，则 $\operatorname{int}\overline{K}=\operatorname{int}K$：闭包内点周围可在 K 选取逼近一组包住该点的小单纯形顶点，凸性使更小邻域在 K，故仍可用前述两情形。必要性：若 x 为内点，沿 p 的正负小位移均在 K，不可能在面同侧。空 K 时任意过 x 的面都满足。

64 原课件第 64 页：支撑超平面的完整证明

Vectors in $\mathbb{R}^N$

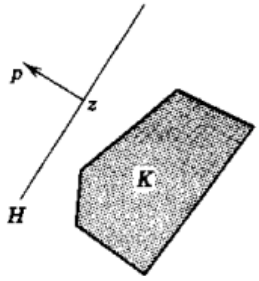


Fig. 7.a

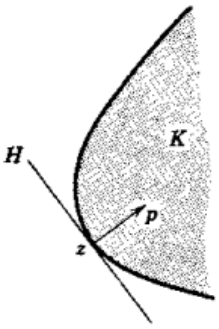


Fig. 7.b

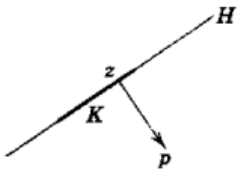


Fig. 7.c

The intuitive content of the result in R^2 is brought out by fig. 7.a, where z is exterior to K, by fig. 7.b, where z is in the boundary of K, and by fig. 7.c, where K (the heavy-lined segment) has no interior. It is easy to draw a non-convex set in R^2 for which some non-interior points are in no bounding hyperplane.

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本页继续“支撑超平面的完整证明”。完整定义、条件和论证见本讲原页第 63 页下方。

65 原课件第 65 页：极锥与坐标序

Vectors in $\mathbb{R}^N$

y. Let C be a cone with vertex 0. Its polar is the set $C^\circ = \{x \in \mathbb{R}^N \mid x \cdot x' \leqq 0 \text{ for every } x' \in C\}$. It is easy to see that C° is a closed, convex cone with vertex 0.
Figure 8 provides an illustration in $\mathbb{R}^2$.

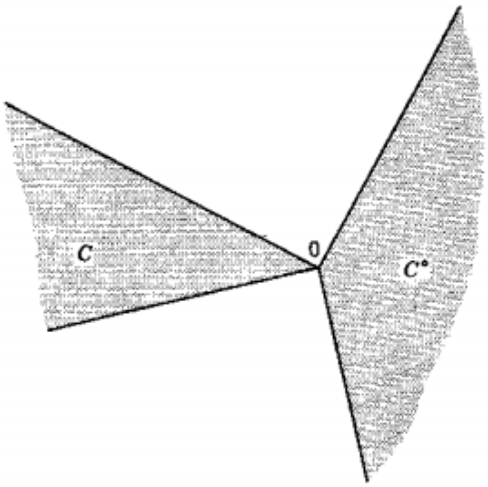


Fig. 8

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$C^\circ = \{p : p \cdot x \leq 0\ \forall x \in C\}$ 是各个闭半空间的交，故闭凸，且非负倍乘仍满足不等式，所以是锥。正正交象限 $\mathbb{R}^N_+$ 的极锥是 $\mathbb{R}^N_-$：取各坐标单位向量证明必要，逐项相乘求和证明充分。坐标序的 $<$ 只要求至少一坐标严格，$\ll$ 要求全部严格。

66 原课件第 66 页：极锥与坐标序



z. According to the general convention of 1.4.j, an ordering is defined on $\mathbb{R}^N$ by $x^1 \leqq x^2$ if $x_n^1 \leqq x_n^2$ for every $n = 1, \cdots, N$. As remarked in 1.4.j, $x^1 < x^2$ means "$x_n^1 \leqq x_n^2$ for all n and $x_n^1 < x_n^2$ for at least one n," and $x^1 \ll x^2$ means "$x_n^1 < x_n^2$ for all n." (Several authors have used the notation $\leqq, \leq, <$ respectively for $\leqq, <, \ll$.)

The non-negative orthant of $\mathbb{R}^N$ is the set $\Omega = \{x \in \mathbb{R}^N \mid x \geqq 0\}$. The letter Ω has throughout this volume the meaning introduced here.

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本页继续“极锥与坐标序”。完整定义、条件和论证见本讲原页第 65 页下方。

67 原课件第 67 页：三种无穷远方向及量词

Definitions

Consider a subset S of $\mathbb{R}^N$. To describe those of its points that are infinitely far from 0 one introduces the concept of its asymptotic cone, as follow. Let M be a positive real number, and denote by S^M the set $\{x \in S \mid \|x\| \geqq M\}$ of vectors in S whose norm is greater than M. Let $\Gamma\left(S^M\right)$ be the least closed cone with vertex 0 containing S^M (i.e., the intersection of all the closed cones with vertex 0 containing S^M). The asymptotic cone of S, denoted $\mathbf{A}S$, is defined as the intersection of all the $\Gamma\left(S^M\right)$, i.e.,

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right);$$

it is clearly a closed cone with vertex 0.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

截尾定义先保留 $\|x\| \geq M$ 的点，再取所生成最小闭锥，最后对全部 M 取交。序列定义为 $d \in C_\infty \iff \exists x_n \in C,\ t_n \to +\infty : x_n/t_n \to d$。衰退锥为 $\operatorname{rec} C = \{d : \forall x \in C, \forall t \geq 0, x + td \in C\}$，要求所有起点都可以一直沿该方向走；只在非空闭凸情形才保证与渐近锥一致。单位方向归一化必须选范数趋于无穷的子列，而不仅是声称任意无界序列范数都趋无穷。截尾最小闭锥不是凸锥，闭包横线不能漏。

68 原课件第 68 页：三种无穷远方向及量词

Definitions

For a given subset of $\mathbb{R}^N$ we are interested in studying its **behavior at infinity**. This leads to the concepts of asymptotic cone. There are three kinds of definition on asymptotic cone.

Definition 1.1. The asymptotic cone of S, denoted $\mathbf{A}S$, is defined as the intersection of all the $\Gamma\left(S^M\right)$,

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right),$$

where $\Gamma\left(S^M\right)$ be the least closed cone with vertex 0 containing $S^M = \{x \in S \mid \|x\| \geqq M\}$ (i.e., the intersection of all the closed cones with vertex 0 containing S^M).

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“三种无穷远方向及量词”。完整定义、条件和论证见本讲原页第 67 页下方。

69 原课件第 69 页：三种无穷远方向及量词

Definitions

Remark. If $M_1 \leqq M_2$, then $S^{M_1} \supseteq S^{M_2}$ and $\Gamma\left(S^{M_1}\right) \supseteq \Gamma\left(S^{M_2}\right)$. $\Gamma\left(S^M\right)$ is a monotonically decreasing set-sequence. Hence

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right) = \lim_{M \to \infty} \Gamma\left(S^M\right).$$

Therefore $\mathbf{A}S$ is the cone generated by the points in S which are at infinity.

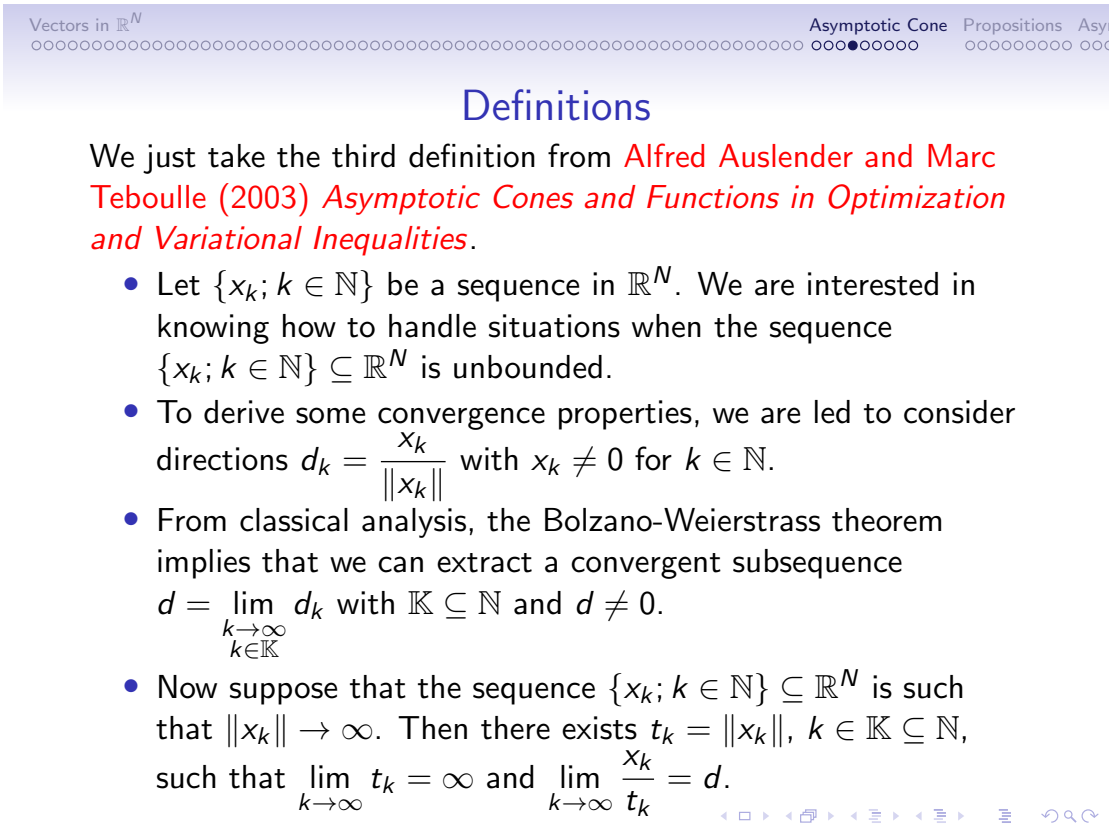
As $\Gamma\left(S^M\right)$ is the least closed cone with vertex 0 containing S_M, then we can see that

$$\begin{aligned} \Gamma\left(S^M\right) &= \overline{\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\}} \\ &= \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S^M}\right\} \\ &= \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S}^M\right\} \end{aligned}$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“三种无穷远方向及量词”。完整定义、条件和论证见本讲原页第 67 页下方。

70 原课件第 70 页：三种无穷远方向及量词



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本页继续“三种无穷远方向及量词”。完整定义、条件和论证见本讲原页第 67 页下方。

71 原课件第 71 页：三种无穷远方向及量词

Definitions

A sequence $\{x_k; k \in \mathbb{N}\} \subseteq \mathbb{R}^N$ is said to converge to a direction $d \in \mathbb{R}^N$ if there exists $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$, with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$.

Definition 1.2. Let C be a non-empty set in $\mathbb{R}^N$. Then the asymptotic cone of the set C, denoted by C_∞, is the set of vectors $d \in \mathbb{R}^N$ that are limits in direction of the sequences $\{x_k; k \in \mathbb{N}\} \subseteq C$, namely

$$C_\infty = \left\{ d \in \mathbb{R}^N \mid \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq C \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \text{ with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \right\}$$

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72 原课件第 72 页：三种无穷远方向及量词



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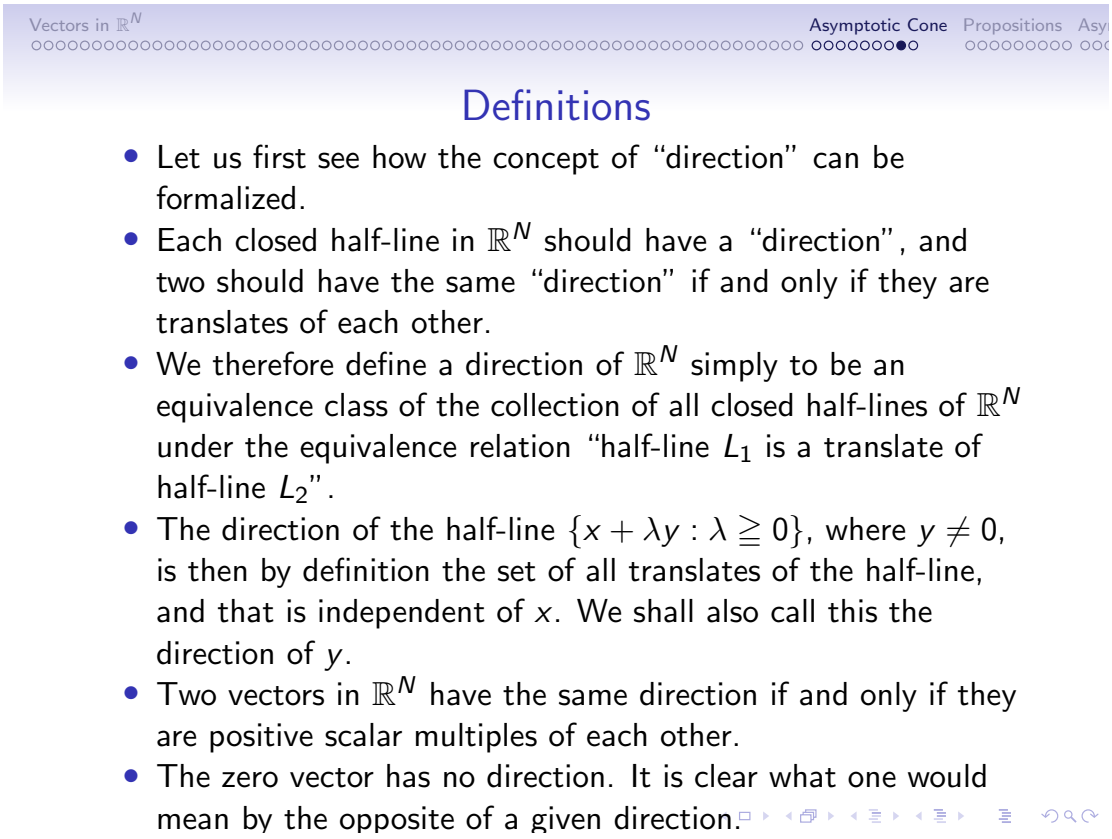
73 原课件第 73 页：三种无穷远方向及量词



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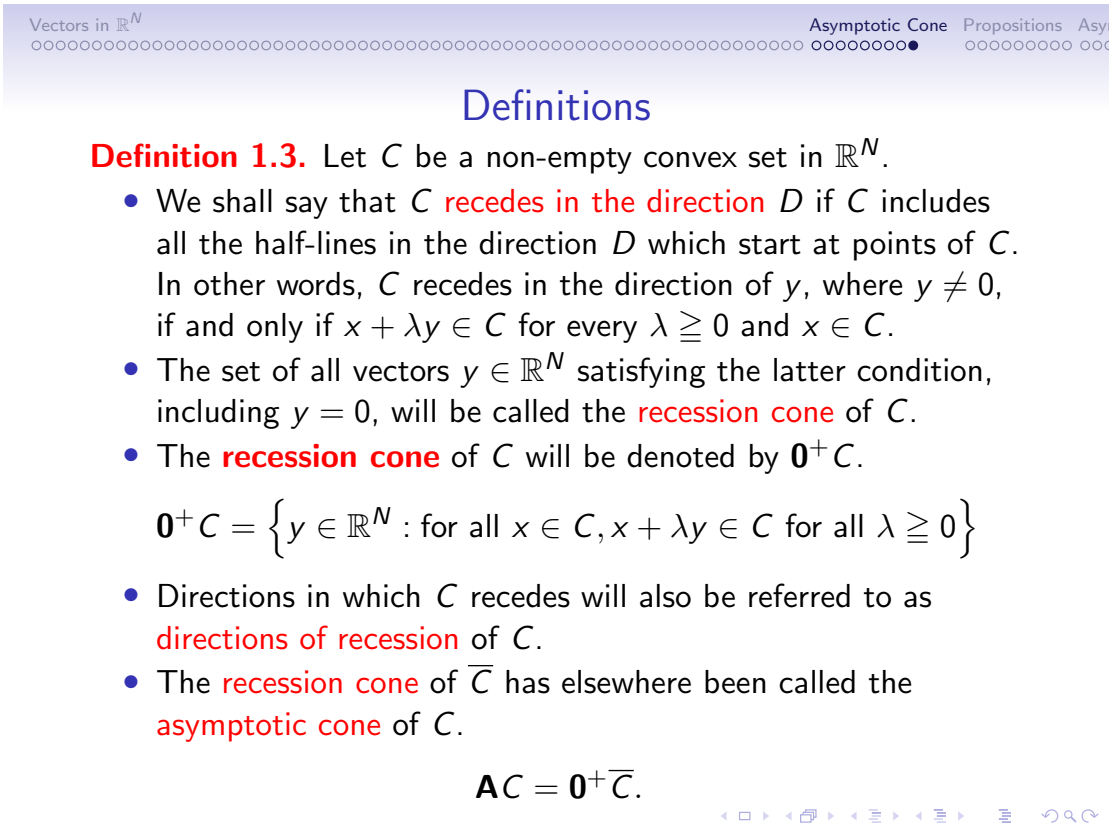
74 原课件第 74 页：三种无穷远方向及量词



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本页继续“三种无穷远方向及量词”。完整定义、条件和论证见本讲原页第 67 页下方。

75 原课件第 75 页：三种无穷远方向及量词



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本页继续“三种无穷远方向及量词”。完整定义、条件和论证见本讲原页第 67 页下方。

76 原课件第 76 页：渐近锥闭性及闭包不变

Proposition of Definition 2

Proposition 2.1. Let $C \subseteq \mathbb{R}^N$ be non-empty.

(1) C_∞ is a closed cone.

(2) $\left(\overline{C}\right)_\infty = C_\infty$.

(3) If C is a cone, then $C_\infty = \overline{C}$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

非空 C 时固定 $x \in C$ 并取 $t_n = n$ 得0属于渐近锥。正倍数通过把 t_n 换为 t_n/λ 得到。闭性用对角选取：若 $d_j \to d$，对每个 j 从其见证序列选一点使 $t_j > j$ 且 $\|x_j/t_j - d_j\| < 1/j$；三角不等式给 $x_j/t_j \to d$。闭包不改变锥：用 $x'_j \in C$ 逼近 $x_j \in \overline{C}$ 至误差 $1/j$，除以 t_j 后误差更趋0。若 C 本身是闭锥，取 $x_n = nd, t_n = n$ 得 $C \subset C_\infty$，反向用每个 $x_n/t_n \in C$ 与闭性；非闭锥的结论为 $C_\infty = \overline{C}$。

77 原课件第 77 页：渐近锥闭性及闭包不变

Proposition of Definition 2

Proof of (1). $0 \in C_\infty$ because if we fix $x_k = x \in C$, for any $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$, we have $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = 0$.

If $d \in C_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. For any $\lambda > 0$, we have $\left\{\dfrac{1}{\lambda}t_k; k \in \mathbb{N}\right\} \subseteq \mathbb{R}$ with $\dfrac{1}{\lambda}t_k \to \infty$, we have $\lim\limits_{k\to\infty} \dfrac{x_k}{\frac{1}{\lambda}t_k} = \lambda d$, i.e. $\lambda d \in C_\infty$.

Hence C_∞ is a cone.

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本页继续“渐近锥闭性及闭包不变”。完整定义、条件和论证见本讲原页第 76 页下方。

78 原课件第 78 页：渐近锥闭性及闭包不变

Proposition of Definition 2

Next, we need to show C_∞ is closed.

For any $d \in \overline{C_\infty}$, there exists $\{d^q; q \in \mathbb{N}\} \subseteq C_\infty$ such that $d^q \to d$. For any d^q, we know there exist $\left\{x_k^q; k \in \mathbb{N}\right\} \subseteq C$ and $\left\{t_k^q; k \in \mathbb{N}\right\} \subseteq \mathbb{R}$ with $\lim\limits_{k\to\infty} t_k^q = \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k^q}{t_k^q} = d^q$.
Then there exists $K_q \in \mathbb{N}$ such that for all $k \geqq K_q$, we have $t_k^q > q$ and $\left\| \dfrac{x_k^q}{t_k^q} - d^q \right\| < \dfrac{1}{q}$. Let $\left\{x_q = x_{K_q}^q\right\} \subseteq C$ and $\left\{t'_q = t_{K_q}^q\right\} \subseteq \mathbb{R}$, we have $t'_q = t_{K_q}^q > q$ hence $\lim\limits_{q\to\infty} t'_q = \infty$.

$$\left\| \frac{x_q}{t'_q} - d \right\| \leqq \left\| \frac{x_q}{t'_q} - d^q \right\| + \|d^q - d\| < \frac{1}{q} + \|d^q - d\| \to 0$$

hence $\lim\limits_{q\to\infty} \dfrac{x_q}{t'_q} = d$, which implies $d \in C_\infty$. Hence C_∞ is a closed cone.

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本页继续“渐近锥闭性及闭包不变”。完整定义、条件和论证见本讲原页第 76 页下方。

79 原课件第 79 页：渐近锥闭性及闭包不变

Proposition of Definition 2

Proof of (2). It's trivial that $C_\infty \subseteq \left(\overline{C}\right)_\infty$. Conversely, let $d \in \left(\overline{C}\right)_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq \overline{C}$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. For any $x_k \in \overline{C}$, there exists $x_k' \in C$ such that $\|x_k' - x_k\| < \dfrac{1}{k}$. Hence

$$\begin{aligned}\lim_{k\to\infty} \left\| \frac{x_k'}{t_k} - d \right\| &\leqq \lim_{k\to\infty} \left[\left\| \frac{x_k'}{t_k} - \frac{x_k}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| \right] \\ &< \lim_{k\to\infty} \frac{1}{k\,|t_k|} + \lim_{k\to\infty} \left\| \frac{x_k}{t_k} - d \right\| \longrightarrow 0\end{aligned}$$

hence $\lim\limits_{k\to\infty} \dfrac{x_k'}{t_k} = d$, i.e. $d \in C_\infty$. Hence $\left(\overline{C}\right)_\infty = C_\infty$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“渐近锥闭性及闭包不变”。完整定义、条件和论证见本讲原页第 76 页下方。

80 原课件第 80 页：渐近锥闭性及闭包不变

Proposition of Definition 2

Proof of (3). We just need to prove that if C is a closed cone, then $C_\infty = C$. Then for any cone C, we have $C_\infty = \left(\overline{C}\right)_\infty = \overline{C}$ as $\overline{C}$ is also a cone.

If C is a closed cone, then for any $x \in C$, for any $k > 0$, we have $kx \in C$. Let $x_k = kx$ and $t_k = k$, we have $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = x$, hence $x \in C_\infty$, i.e. $C \subseteq C_\infty$.

For any $d \in C_\infty$, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. Note that C is a cone, $\dfrac{x_k}{t_k} \in C$, then $d \in \overline{C} = C$. Hence $C_\infty = C$. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“渐近锥闭性及闭包不变”。完整定义、条件和论证见本讲原页第 76 页下方。

81 原课件第 81 页：截尾与序列定义等价

Proposition of Definition 2

Theorem 2.1. If S is a subset of $\mathbb{R}^N$, then

$$\mathbf{A}S = S_\infty$$

where

$$\mathbf{A}S = \bigcap_{k\geqq 0} \Gamma\left(S_M\right)$$

with $S^M = \{x \in S \mid \|x\| \geqq M\}$ and
$\Gamma\left(S^M\right) = \{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\}$ and

$$S_\infty = \left\{ d \in \mathbb{R}^N \mid \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq S \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \right\}.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

非零 $d = x_n/t_n$ 的极限使 $\|x_n\| \to \infty$，所以对任意 M，最终原点均属截尾集，其缩放极限落入截尾闭锥，故属于交。反向对非零 d，每个整数 M 选 $\alpha_M x_M$ 距 d 小于 $1/M$，且 $\|x_M\| \geq M$。由 $\alpha_M \leq (\|d\| + 1/M)/M \to 0$，取 $t_M = 1/\alpha_M \to \infty$，得到见证序列。零向量对非空集合单独处理。归一化方向全体再取非负倍数得到相同锥；有界集方向集为空时需单独按约定加入0。

82 原课件第 82 页：截尾与序列定义等价

Proposition of Definition 2

Proof. For any $d \in S_\infty$. If $d = 0$, then $d \in \mathbf{A}S$ is trivial. If $d \neq 0$, then there exist $\{x_n\} \subseteq S$ and $\{t_n\} \subseteq \mathbb{R}$ with $t_n \to \infty$ such that $\lim_{n\to\infty} \frac{x_n}{t_n} = d$. There exists $N \in \mathbb{N}$ such that for all $n \geqq N$, we have $\left\| \frac{x_n}{t_n} \right\| > \frac{\|d\|}{2} > 0$. Then for any $M \geqq 0$, as $t_n \to \infty$, there exists $N_M > 0$ such that for all $n \geqq N_M$, we have $\|x_n\| \geqq M$. Hence $x_n \in S^M$. Hence $\frac{x_n}{t_n} \in \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\right\}$ for all $n \geqq N_M$. As $\lim_{n\to\infty} \frac{x_n}{t_n} = d$, we have $d \in \Gamma\left(S^M\right)$ for any $M \geqq 0$. Thus,

$$S_\infty \subseteq \bigcap_{M\geqq 0} \Gamma\left(S^M\right) = \mathbf{A}S$$

Conversely, if $d \in \mathbf{A}S$ with $d \neq 0$, then for all $M \geqq 0$, $d \in \Gamma\left(S^M\right) = \overline{\left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\right\}}$, so there exists $\alpha_M x_M$ where $\alpha_M > 0$ and $x_M \in S^M$ such that $\|d - \alpha_M x_M\| < \frac{1}{M}$. Then let $t_M = \frac{1}{\alpha_M}$, if we can prove $t_M \to \infty$, i.e. $\alpha_M \to 0$, then we are done. As $\|\alpha_M x_M\| \leqq \|d\| + \|d - \alpha_M x_M\|$, then

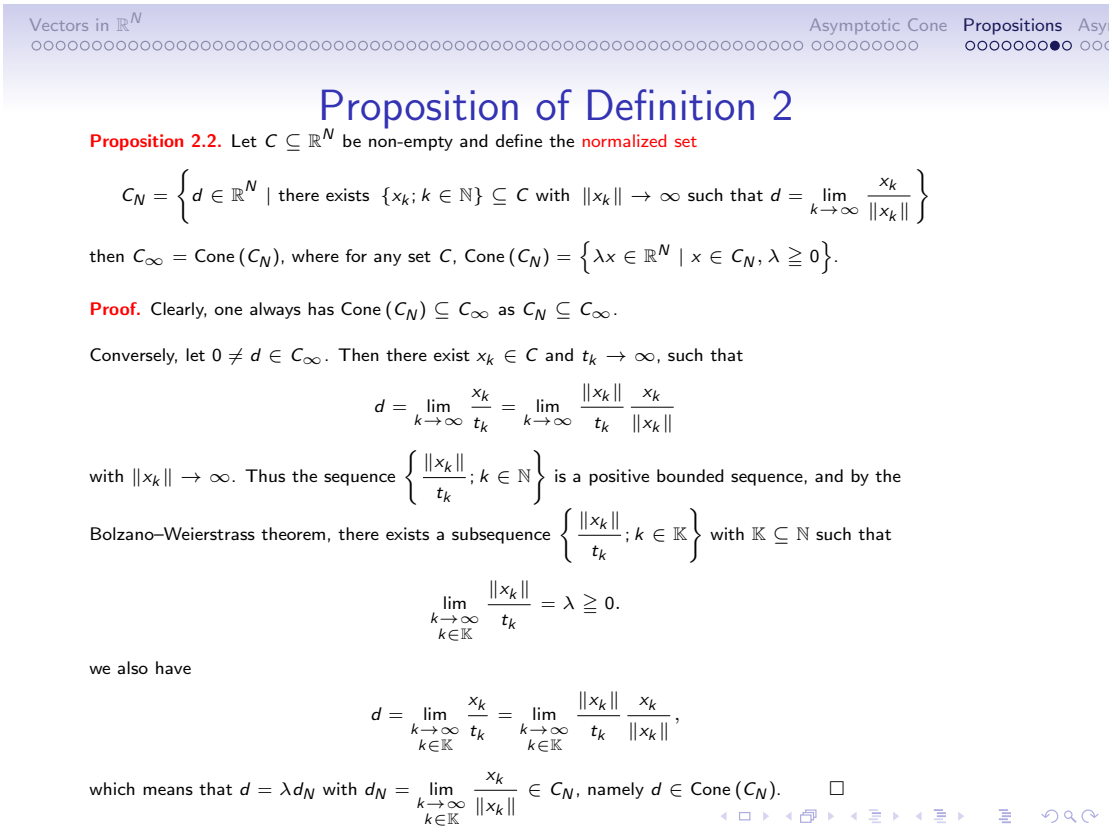
$$|\alpha_M| \leqq \frac{\|d\| + \|d - \alpha_M x_M\|}{\|x_M\|} < \frac{\|d\| + \frac{1}{M}}{M} \to 0.$$

We are done! □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“截尾与序列定义等价”。完整定义、条件和论证见本讲原页第 81 页下方。

83 原课件第 83 页：截尾与序列定义等价



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“截尾与序列定义等价”。完整定义、条件和论证见本讲原页第 81 页下方。

84 原课件第 84 页：有界判据与平移不变

Proposition of Definition 2

Proposition 2.3. A set $C \subseteq \mathbb{R}^N$ is bounded if and only if $C_\infty = \{0\}$.

Proof. It is clear that C_∞ cannot contain any non-zero direction if C is bounded. As C is bounded, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = 0$. Hence $C_\infty = \{0\}$.

Conversely, if C is unbounded, then there exists a sequence $\{x_k; k \in \mathbb{N}\} \subseteq C$ with $x_k \neq 0$ for all $k \in \mathbb{N}$ such that $\|x_k\| \to \infty$ and thus the vectors $d_k = \frac{x_k}{\|x_k\|} \in \{d \in \mathbb{R}^N : \|d\| = 1\}$.
Therefore, we can extract a subsequence of $\{d_k\}$ such that $\lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} d_k = d$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|d\| = 1$. This non-zero vector d is an element of C_∞, a contradiction! □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

非空有界 C 的任意 x_n/t_n 趋于0，故渐近锥只有0。若无界，选 $\|x_n\| \to \infty$，单位向量抽子列趋于单位 d，令 $t_n = \|x_n\|$ 得非零渐近方向，矛盾。平移 $C + a$ 不改变方向，因为 $(x_n + a)/t_n - x_n/t_n = a/t_n \to 0$。若 E 非空，固定 $a \in E$，由 $C + a \subset C + E$ 及单调性得 $C_\infty \subset (C + E)_\infty$。

85 原课件第 85 页：有界判据与平移不变

Asymptotic Cone

Theorem 3.1. If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S+\{x\}) = \mathbf{A}S$.

Proof. We just need to prove $\mathbf{A}(S+\{x\}) \subseteq \mathbf{A}S$, then to prove $\mathbf{A}S \subseteq \mathbf{A}(S+\{x\})$, we just need to let $S = (S+\{x\}) + \{-x\}$ and we arrive at the conclusion. Let $T = S + \{x\}$.

For any $d \in \mathbf{A}(S+\{x\})$, $d \in (S+\{x\})_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq S+\{x\}$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. Let $\{x'_k = x_k - x\} \subseteq S$ and

$$\left\| \frac{x'_k}{t_k} - d \right\| \leqq \left\| \frac{x'_k}{t_k} - \frac{x_k}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| = \left\| \frac{x}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| \to 0$$

hence $\lim\limits_{k\to\infty} \dfrac{x'_k}{t_k} = d$, i.e., $d \in S_\infty$, hence $d \in \mathbf{A}S$. Hence $\mathbf{A}(S+\{x\}) \subseteq \mathbf{A}S$. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“有界判据与平移不变”。完整定义、条件和论证见本讲原页第 84 页下方。

86 原课件第 86 页：有界判据与平移不变

Asymptotic Cone

Theorem 3.1. If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S + \{x\}) = \mathbf{A}S$.

Proof.

$$
\begin{aligned}
& d \in \mathbf{A}(S) \Longleftrightarrow d \in S_\infty \\
\Longleftrightarrow \quad & \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq S \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \\
\Longleftrightarrow \quad & \text{there exist } \{x_k + x; k \in \mathbb{N}\} \subseteq S + \{x\} \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k + x}{t_k} = d \\
\Longleftrightarrow \quad & d \in (S + \{x\})_\infty \Longleftrightarrow d \in \mathbf{A}(S + \{x\})
\end{aligned}
$$

□

In other words, a translation of S does not alter its asymptotic cone.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“有界判据与平移不变”。完整定义、条件和论证见本讲原页第 84 页下方。

87 原课件第 87 页：有界判据与平移不变

Asymptotic Cone

Theorem 3.2. If $E \neq \varnothing$ and S are two subsets of $\mathbb{R}^N$, then $\mathbf{A}S \subseteq \mathbf{A}(S + E)$.

Proof. Let $x \in E$. Then $\mathbf{A}S = \mathbf{A}(S + \{x\}) \subseteq \mathbf{A}(S + E)$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“有界判据与平移不变”。完整定义、条件和论证见本讲原页第 84 页下方。

88 原课件第 88 页：乘积与交集的渐近方向

Asymptotic Cone

Theorem 3.3. If, for every $n = 1, \cdots, N$, S_n is a subset of $\mathbb{R}^{K_n}$, then $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$.

Proof. For $d \in \mathbf{A}\left(\prod_{n=1}^{N} S_n\right)$. If $d = 0$, then $d \in \prod_{n=1}^{N} \mathbf{A}S_n$ is trivial. If $d \neq 0$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq \prod_{n=1}^{N} S_n$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. Let

$$x_k = \left(x_k^1, \cdots, x_k^n, \cdots, x_k^N\right), d = \left(d^1, \cdots, d^n, \cdots, d^N\right)$$

where $x_k^n \in S_n$ and $d^n \in \mathbb{R}$ for $n = 1, \cdots, N$. For each $n = 1, \cdots, N$, we have $\lim_{k\to\infty} \frac{x_k^n}{t_k} = d^n$. Hence $d^n \in \mathbf{A}S_n$, hence $d \in \prod_{n=1}^{N} \mathbf{A}S_n$. Thus, $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $(x_{1n}, \ldots, x_{mn})/t_n \to (d_1, \ldots, d_m)$，逐坐标就是 $x_{in}/t_n \to d_i$，所以积的渐近锥包含于渐近锥之积。反向一般不能单靠各自见证序列，因为缩放速度未必相同。第89页把总范数截尾当成各分量都截尾的等式错误，例如 $(M, 0)$ 总范数大但第二分量不大。交集包含于每个集合，锥的单调性给交集锥包含于各锥交；若右边只有0，交集有界（空交集本身有界，非空时用第84页）。

89 原课件第 89 页：乘积与交集的渐近方向

Asymptotic Cone

Theorem 3.3. If, for every $n = 1, \cdots, N$, S_n is a subset of $\mathbb{R}^{K_n}$, then $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$.

Proof. $y \in \mathbf{A}\left(\prod_{n=1}^{N} S_n\right)$, then, for any $M \geqq 0$, $y \in \Gamma\left(\left(\prod_{n=1}^{N} S_n\right)^M\right)$, then

$$
\begin{aligned}
y \quad &\in \quad \overline{\left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \left(\prod_{n=1}^{N} S_n\right)^M\right\}} = \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{\prod_{n=1}^{N} S_n^M}\right\} \\
&\subseteq \quad \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \prod_{n=1}^{N} \overline{S_n^M}\right\} = \prod_{n=1}^{N} \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S_n^M}\right\} \\
&= \quad \prod_{n=1}^{N} \overline{\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S_n^M\}} = \prod_{n=1}^{N} \Gamma\left(S_n^M\right)
\end{aligned}
$$

hence $y \in \bigcap_{M \geqq 0}\left[\prod_{n=1}^{N} \Gamma\left(S_n^M\right)\right] = \prod_{n=1}^{N}\left[\bigcap_{M \geqq 0} \Gamma\left(S_n^M\right)\right] = \prod_{n=1}^{N} \mathbf{A}(S_n)$. Therefore $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$.

□

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“乘积与交集的渐近方向”。完整定义、条件和论证见本讲原页第 88 页下方。

90 原课件第 90 页：乘积与交集的渐近方向

Asymptotic Cone

Theorem 3.4. Given a collection of subsets of $\mathbb{R}^N$, if the intersection of their asymptotic cones is $\{0\}$, then their intersection is bounded.

Proof. As we have that "$S_1 \subseteq S_2$" implies "$\mathbf{A}S_1 \subseteq \mathbf{A}S_2$", then, for an arbitrary index set Γ, let $C_\gamma \subseteq \mathbb{R}^N$, $\gamma \in \Gamma$, we have

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \mathbf{A}C_\gamma$$

for all $\gamma \in \Gamma$. Hence

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma$$

If $\bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma = \{0\}$, then $\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) = \{0\}$. By Proposition 2.3, we have $\bigcap_{\gamma\in\Gamma} C_\gamma$ is bounded. □

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本页继续“乘积与交集的渐近方向”。完整定义、条件和论证见本讲原页第 88 页下方。

91 原课件第 91 页：乘积与交集的渐近方向

Asymptotic Cone

Theorem 3.4. For all $j \in \mathcal{J}$, $S_j \subseteq \mathbb{R}^N$, if $\bigcap_{j\in\mathcal{J}} \mathbf{A}(S_j) = \{0\}$, then $\bigcap_{j\in\mathcal{J}} S_j$ is bounded.

Proof. For all $j \in \mathcal{J}$, $\bigcap_{j\in\mathcal{J}} S_j \subseteq S_j$, then $\mathbf{A}\left(\bigcap_{j\in\mathcal{J}} S_j\right) \subseteq \mathbf{A}(S_j)$. Whence $\mathbf{A}\left(\bigcap_{j\in\mathcal{J}} S_j\right) \subseteq \bigcap_{j\in\mathcal{J}} \mathbf{A}(S_j) = \{0\}$. There exists $M_0 \geqq 0$ such that $\left(\bigcap_{j\in\mathcal{J}} S_j\right)^M = \varnothing$ for $M \geqq M_0$, thus $\bigcap_{j\in\mathcal{J}} S_j$ is bounded. □

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本页继续“乘积与交集的渐近方向”。完整定义、条件和论证见本讲原页第 88 页下方。

92 原课件第 92 页：线性像定理再证明

Asymptotic Cone

Theorem 3.5. (Images under Linear Mappings) For a linear mapping $L : \mathbb{R}^N \to \mathbb{R}^K$ and a closed set $C \subseteq \mathbb{R}^N$, a sufficient condition for the set $L(C)$ to be closed is that $L^{-1}\{0\} \cap C_\infty = \{0\}$. Under this condition one has $L(C_\infty) = L(C)_\infty$, although in general only $L(C_\infty) \subseteq L(C)_\infty$.

Theorem 3.5. (Images under Linear Mappings) For a linear mapping $L : \mathbb{R}^N \to \mathbb{R}^K$ and a closed set $C \subseteq \mathbb{R}^N$, if $L^{-1}\{0\} \cap C_\infty = \{0\}$, then

(1) the set $L(C)$ is closed;

(2) $L(C_\infty) = L(C)_\infty$.

Although in general only $L(C_\infty) \subseteq L(C)_\infty$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本段重复第37–41页的重要定理。核心核条件为 $\ker L \cap C_\infty = \{0\}$。若 Lx_n 有界而 x_n 无界，取范数趋无穷子列并单位化，极限是非零渐近方向且被 L 送到0，矛盾。因此像序列收敛时原像可抽收敛子列，闭 C 与连续 L 给像闭。锥像的容易包含是对 x_n/t_n 直接取 L；困难包含须用第41页的缩放反证，不可仅引用原集合上的有界性。

93 原课件第 93 页：线性像定理再证明

Asymptotic Cone

Proof. The condition will be shown first to imply that L carries unbounded sequences in C into unbounded sequences in $\mathbb{R}^K$. If this were not true, there would exist a sequence of points $\{x_k; k \in \mathbb{N}\} \subseteq C$ such that $\|x_k\| \to \infty$, but such that the sequence of images $y_k = L(x_k)$ is bounded. As $\left\{ \frac{x_k}{\|x_k\|}; k \in \mathbb{N} \right\}$ is bounded, so we can extract a subsequence of $\left\{ \frac{x_k}{\|x_k\|}; k \in \mathbb{K} \right\}$ such that $\lim\limits_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{x_k}{\|x_k\|} = x$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|x\| = 1$. Hence $x \in C_\infty$, but also (because linear mappings are continuous)

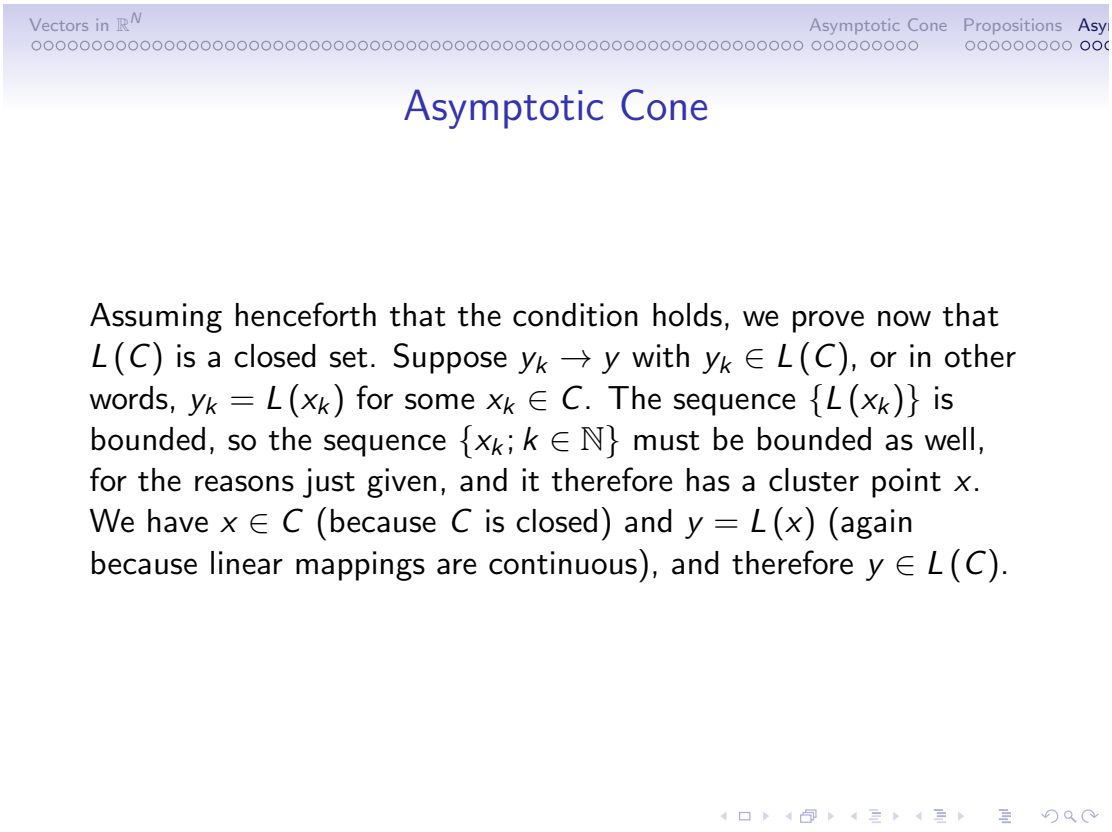
$$L(x) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{1}{\|x_k\|} L(x_k) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{y_k}{\|x_k\|} = 0$$

which the condition prohibits.

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本页继续“线性像定理再证明”。完整定义、条件和论证见本讲原页第 92 页下方。

94 原课件第 94 页：线性像定理再证明



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本页继续“线性像定理再证明”。完整定义、条件和论证见本讲原页第 92 页下方。

95 原课件第 95 页：线性像定理再证明

Asymptotic Cone

The next task is to show that $L(C_\infty) \subseteq L(C)_\infty$. This is trivial when $C = \varnothing$, so we assume $C \neq \varnothing$. Any $d \in C_\infty$, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. The points $y_k = L(x_k)$ in $L(C)$ and

$$\lim_{k\to\infty} \frac{y_k}{t_k} = \lim_{k\to\infty} \frac{L(x_k)}{t_k} = L\left(\lim_{k\to\infty} \frac{x_k}{t_k}\right) = L(d)$$

hence $L(d) \in L(C)_\infty$. Hence $L(C_\infty) \subseteq L(C)_\infty$.

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本页继续“线性像定理再证明”。完整定义、条件和论证见本讲原页第 92 页下方。

96 原课件第 96 页：线性像定理再证明

Asymptotic Cone

For the inclusion $L(C_\infty) \supseteq L(C)_\infty$, we consider any $y \in L(C)_\infty$, there exist $\{y_k; k \in \mathbb{N}\} \subseteq L(C)$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{y_k}{t_k} = y$. For any choice of $x_k \in C$ with $L(x_k) = y_k$, we have $\lim_{k\to\infty} L\left(\frac{x_k}{t_k}\right) = y$. The boundedness of the sequence $\left\{L\left(\frac{x_k}{t_k}\right); k \in \mathbb{N}\right\}$ implies the boundedness of the sequence $\left\{\frac{x_k}{t_k}; k \in \mathbb{N}\right\}$, which therefore must have a cluster point x. Then $x \in C_\infty$ and $L(x) = y$, and we conclude that $y \in L(C_\infty)$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“线性像定理再证明”。完整定义、条件和论证见本讲原页第 92 页下方。

97 原课件第 97 页：闭和定理的两个证明

Asymptotic Cone

Theorem 3.6 (Sums of Sets). For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, a sufficient condition for $\sum_{n=1}^{N} C_n$ to be closed is the non-existence of vectors $x_n \in (C_n)_\infty$ such that $\sum_{n=1}^{N} x_n = 0$, except with $x_n = 0$ for all n. Then also

$$\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty .$$

Theorem 3.6 (Sums of Sets). For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, if $(C_1)_\infty, \cdots, (C_N)_\infty$ are positively semi-independent (P.S.I.), then

(1) $\sum_{n=1}^{N} C_n$ is closed;

(2) $\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty$.

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积集加求和映射的证明已在第42页完整给出。两集合的直接证明：若 $z_n = a_n + b_n \to z$ 而 a_n 无界，选 $r_n = \|a_n\| \to \infty$ 及 $a_n/r_n \to d$，$\|d\| = 1$。由 $z_n/r_n \to 0$ 得 $b_n/r_n \to -d$，于是 $d \in A_\infty, -d \in B_\infty$，违反正半独立。故 a_n 有界，抽子列趋于 $a \in A$，再由 $b_n = z_n - a_n \to z - a \in B$，得 $z \in A + B$。有限多集合用积方法最直接，避免在归纳时漏检中间和的渐近锥条件。

98 原课件第 98 页：闭和定理的两个证明

Asymptotic Cone

Proof. Let $C = \prod_{n=1}^{N} C_n$. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n$$

for $x_n \in C_n$. Then L is a linear mapping. As Theorem 3.5 shows, $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$. And by the condition, we have

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}$$

hence

$$L^{-1}(0) \cap C_\infty = \{0\}$$

so by Theorem 3.5, we have $L(C)$ is closed. As $L(C) = \sum_{n=1}^{N} C_n$, we have that $\sum_{n=1}^{N} C_n$ is closed.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭和定理的两个证明”。完整定义、条件和论证见本讲原页第 97 页下方。

99 原课件第 99 页：闭和定理的两个证明

Asymptotic Cone

And again by Theorem 3.5, $L(C_\infty) = L(C)_\infty$, i.e.

$$L\left(\left(\prod_{n=1}^{N} C_n\right)_\infty\right) = \left(\sum_{n=1}^{N} C_n\right)_\infty$$

We also have

$$L\left(\prod_{n=1}^{N} (C_n)_\infty\right) = \sum_{n=1}^{N} (C_n)_\infty$$

and

$$\left(\prod_{n=1}^{N} C_n\right)_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$$

hence

$$\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty$$

□

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本页继续“闭和定理的两个证明”。完整定义、条件和论证见本讲原页第 97 页下方。

100 原课件第 100 页：闭和定理的两个证明

Asymptotic Cone

Proof. Let $C = \prod_{n=1}^{N} C_n$. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n$$

for $x_n \in C_n$. Then L is a linear mapping. As Theorem 3.5 shows, $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$. And by the condition, we have

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}.$$

$$(x_1, \cdots, x_N) \in L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty$$

$$L((x_1, \cdots, x_N)) = 0 \quad \text{and} \quad x_n \in (C_n)_\infty \ \text{ for } n = 1, \cdots, N$$

$$x_n \in (C_n)_\infty \ \text{ for } n = 1, \cdots, N \quad \text{and} \quad \sum_{n=1}^{N} x_n = 0$$

If $(C_1)_\infty, \cdots, (C_N)_\infty$ are positively semi-independent (P.S.I.), then $x_n = 0$ for $n = 1, \cdots, N$, and hence

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}.$$

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本页继续“闭和定理的两个证明”。完整定义、条件和论证见本讲原页第 97 页下方。

101 原课件第 101 页：闭和定理的两个证明

Asymptotic Cone

Theorem 3.6. $S_1, \cdots, S_j, \cdots, S_J$ are closed subsets of $\mathbb{R}^N$, if their asymptotic cones $\mathbf{A}(S_1), \cdots, \mathbf{A}(S_j), \cdots, \mathbf{A}(S_J)$ are positively semi-independent, then their sum $\sum_{j=1}^{J} S_j$ is closed.

Proof. We first verify the setting of $J = 2$. $\mathbf{A}(S_1)$ and $\mathbf{A}(S_2)$ are positively semi-independent if and only if $\mathbf{A}(S_1) \cap \mathbf{A}(-S_2) = \{0\}$.

Suppose $\mathbf{A}(S_1) \neq \{0\} \neq \mathbf{A}(S_2)$, then S_1 and S_2 are unbounded. We assume that $\{z^K; K \in \mathbb{R}_+\} \subseteq S_1 + S_2$ and $z^K \to z^0$. According to the definition, for any $K \in \mathbb{R}_+$, there exist $x_1^K \in S_1$ and $x_2^K \in S_2$ such that $z^K = x_1^K + x_2^K$.

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本页继续“闭和定理的两个证明”。完整定义、条件和论证见本讲原页第 97 页下方。

102 原课件第 102 页：闭和定理的两个证明

Asymptotic Cone

If $\{x_1^K; K \in \mathbb{R}_+\}$ is unbounded in S_1, then for any $K \in \mathbb{R}_+$, $x_1^K \in S_1^K$, $x_1^K \in \Gamma(S_1^K)$, $\|x_1^K\| \to \infty$, then $x_2^K = z^K - x_1^K$ and $\|x_2^K\| \to \infty$. For any $K \in \mathbb{R}_+$, $-x_2^K \in -S_2^K$, $-x_2^K \in \Gamma(-S_2^K)$, and

$$\frac{x_1^K}{\|x_1^K\|} = \frac{z^K}{\|x_1^K\|} - \frac{x_2^K}{\|x_1^K\|}, \quad \frac{x_1^K}{\|x_1^K\|} \in \Gamma(S_1^K) \quad \text{and} \quad -\frac{x_2^K}{\|x_1^K\|} \in \Gamma(-S_2^K)$$

then

$$\lim_{K\to\infty} \frac{x_1^K}{\|x_1^K\|} = \lim_{K\to\infty} \frac{z^K}{\|x_1^K\|} + \lim_{K\to\infty} -\frac{x_2^K}{\|x_1^K\|},$$

$$\lim_{K\to\infty} \frac{x_1^K}{\|x_1^K\|} \in \Gamma(S_1^K) \qquad \text{and} \qquad \lim_{K\to\infty} -\frac{x_2^K}{\|x_1^K\|} \in \Gamma(-S_2^K)$$

that is,

$$\lim_{K\to\infty} \frac{x_1^K}{\|x_1^K\|} = \lim_{K\to\infty} -\frac{x_2^K}{\|x_1^K\|} = e \neq 0, \quad e \in \Gamma(S_1^K) \quad \text{and} \quad e \in \Gamma(-S_2^K)$$

Hence $0 \neq e \in \Gamma(S_1^K) \cap \Gamma(-S_2^K) = \{0\}$, contradiction!

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭和定理的两个证明”。完整定义、条件和论证见本讲原页第 97 页下方。

103 原课件第 103 页：闭和定理的两个证明

Asymptotic Cone

Therefore $\{x_1^K; K \in \mathbb{R}_+\}$ id bounded in S_1, then there exists a convergent subsequence $\{x_1^{K_n}; K_n \in \mathbb{R}_+\} \subseteq \{x_1^K; K \in \mathbb{R}_+\}$ such that $x_1^{K_n} \to x_1^0 \in S_1$. On the other hand, for any $K_n \in \mathbb{R}_+$, $x_2^{K_n} = z^{K_n} - x_1^{K_n}$, then $x_2^{K_n} \to x_2^0 \in S_2$, and hence $z^{K_n} \to x_1^0 + x_2^0 \in S_1 + S_2$. Thus $z^0 = x_1^0 + x_2^0 \in S_1 + S_2$, and hence $S_1 + S_2$ is closed.

The general conclusion holds from mathematical induction. □

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本页继续“闭和定理的两个证明”。完整定义、条件和论证见本讲原页第 97 页下方。

104 原课件第 104 页：渐近正则性与凸渐近锥

Proposition of Definition 3

Associated with the asymptotic cone C_∞ is the following related concept, which will help us in simplifying the definition of C_∞ in the particular case where $C \subseteq \mathbb{R}^N$ is assumed convex.

Definition Let $C \subseteq \mathbb{R}^N$ be non-empty and define

$$C_\infty^1 = \Big\{ d \in \mathbb{R}^N : \text{for all } t_k \to \infty \text{ there exists } \{x_k; k \in \mathbb{N}\} \subseteq C \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \Big\}$$

We say that C is asymptotically regular if $C_\infty = C_\infty^1$.

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正则性把“存在缩放速度”增强为“每一种趋无穷速度都可匹配”。设 $x_j/s_j \to d$，固定 $a \in C$，给任意 $t_n \to \infty$ 选 $j(n) \to \infty$ 且 $s_{j(n)} \geq t_n$。由凸性 $z_n = a + (t_n/s_{j(n)})(x_{j(n)} - a) \in C$，且 $z_n/t_n = a/t_n + (x_{j(n)} - a)/s_{j(n)} \to d$。因此任意非空凸集渐近正则。对两方向使用共同 $t_n = n$ 的见证点，再作相同权重凸组合，就证明渐近锥凸；其闭锥性质已由第76页给出。

105 原课件第 105 页：渐近正则性与凸渐近锥

Proposition of Definition 3

Proposition 4.1. Let C be a non-empty convex set in $\mathbb{R}^N$. Then C is asymptotically regular.

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本页继续“渐近正则性与凸渐近锥”。完整定义、条件和论证见本讲原页第 104 页下方。

106 原课件第 106 页：渐近正则性与凸渐近锥

Proposition of Definition 3

Proof. The inclusion $C_\infty^1 \subseteq C_\infty$ clearly holds from the definitions of C_∞^1 and C_∞, respectively.

Let $d \in C_\infty$. Then there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t'_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t'_k \to \infty$ such that $d = \lim_{k\to\infty} \frac{x_k}{t'_k}$. Let $x \in C$ and define

$$d_k = \frac{1}{t'_k}(x_k - x),$$

then we have

$$d = \lim_{k\to\infty} d_k$$

and $x + t'_k d_k = x_k \in C$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“渐近正则性与凸渐近锥”。完整定义、条件和论证见本讲原页第 104 页下方。

107 原课件第 107 页：渐近正则性与凸渐近锥

Proposition of Definition 3

Now let $\{t_k; k \in \mathbb{N}\}$ be an arbitrary sequence such that $\lim_{k\to\infty} t_k = \infty$. For any fixed $N \in \mathbb{N}$, there exists $k(N)$ with $\lim_{N\to\infty} k(N) = \infty$ such that $t_N \leqq t'_{k(N)}$, and since C is convex, we have

$$z_N = x + t_N d_{k(N)} \in C.$$

As z_N is on the closed segment $\left[x, x + t'_{k(N)} d_{k(N)}\right]$. Hence,

$$d = \lim_{N\to\infty} \frac{z_N}{t_N},$$

showing that $d \in C^1_\infty$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“渐近正则性与凸渐近锥”。完整定义、条件和论证见本讲原页第 104 页下方。

108 原课件第 108 页：渐近正则性与凸渐近锥

Proposition of Definition 3

Proposition 4.2. Let C be a non-empty convex set in $\mathbb{R}^N$. Then the asymptotic cone C_∞ is a closed convex cone.

Proof. By Proposition 2.1, we have shown that C_∞ is a closed cone. We just need to verify C_∞ is convex.

By Proposition 4.1, we know that for any d_1 and d_2 in C_∞^1, for any $t_k \to \infty$, there exists x_k^1 and x_k^2 in C such that

$$\lim_{k\to\infty} \frac{x_k^1}{t_k} = d_1 \qquad \text{and} \qquad \lim_{k\to\infty} \frac{x_k^2}{t_k} = d_2$$

Then for any $\lambda \in [0,1]$, we have $\lambda x_k^1 + (1-\lambda)\, x_k^2 \in C$ and

$$\lim_{k\to\infty} \frac{\lambda x_k^1 + (1-\lambda)\, x_k^2}{t_k} = \lambda d_1 + (1-\lambda)\, d_2$$

hence $\lambda d_1 + (1-\lambda)\, d_2 \in C_\infty^1$. Hence C_∞^1 is convex, hence C_∞ is convex. $\square$

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本页继续“渐近正则性与凸渐近锥”。完整定义、条件和论证见本讲原页第 104 页下方。

109 原课件第 109 页：衰退锥相等与非闭反例

Proposition of Definition 3

Proposition 4.3. If C is a non-empty subset of $\mathbb{R}^N$, then

$$\mathbf{0}^+C \subseteq C_\infty$$

where

$$\mathbf{0}^+C = \left\{ y \in \mathbb{R}^N : \text{for all } x \in C, x + \lambda y \in C \text{ for all } \lambda \geqq 0 \right\}$$

and

$$C_\infty = \left\{ d \in \mathbb{R}^N \mid \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq C \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \right.$$
$$\left. \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \right\}$$

If C is non-empty closed and convex, then

$$\mathbf{0}^+C = C_\infty.$$

So if C is non-empty convex, then

$$\mathbf{0}^+\overline{C} = C_\infty.$$

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任意非空集合有 $\operatorname{rec} C \subset C_\infty$：固定 $a \in C$，用 $x_n = a + nd$。反向需要闭凸：若 $x_n/t_n \to d$，对任意 $a \in C$ 和固定 $t > 0$，充分大时 $a + (t/t_n)(x_n - a) \in C$，取极限得 $a + td \in C$。所以 $\operatorname{rec} C = C_\infty$。第112页反例 $C = ([0,1) \times [0,\infty)) \cup \{(1,0)\}$ 中，衰退方向必须让 $(1,0)$ 及任意 $(0,y)$ 都留在集合，迫使两个方向分量均0，故衰退锥是 $\{0\}$，不是空集；渐近锥是 $\{(0,v) : v \geq 0\}$，不是整条竖直线。闭凸时 $C + C_\infty = C$ 由衰退定义及0在锥中立即得到。

110 原课件第 110 页：衰退锥相等与非闭反例

Proposition of Definition 3

Proof. For any $y \in \mathbf{0}^+C$, for any $k > 0$, let $t_k = k \to \infty$ and $x_k = x + ky$ for any $x \in C$. Then

$$\left\| \frac{x_k}{t_k} - y \right\| = \left\| \frac{x}{k} \right\| \to 0$$

hence $\lim_{k\to\infty} \frac{x_k}{t_k} = y$, which implies that $\mathbf{0}^+C \subseteq C_\infty$.

If C is non-empty, closed and convex, then for any $d \in C_\infty$, we need to show that for all $x \in C$ and $t > 0$, we have $x + td \in C$. As there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. Let $d_k = \frac{x_k - x}{t_k}$, then we have $x_k = x + t_k d_k \in C$ and

$$\lim_{k\to\infty} \frac{x_k}{t_k} = \lim_{k\to\infty} \frac{x + t_k d_k}{t_k} = \lim_{k\to\infty} \left(d_k + \frac{x}{t_k} \right) = d$$

hence

$$\lim_{k\to\infty} d_k = d.$$

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111 原课件第 111 页：衰退锥相等与非闭反例

Proposition of Definition 3

Take k sufficiently large such that $t \leqq t_k$, then since C is convex, we have

$$x + td_k = \left(1 - \frac{t}{t_k}\right) x + \frac{t}{t_k}\left(x + t_k d_k\right) \in C.$$

Passing to the limit, we thus have $x + td = \lim\limits_{k\to\infty}\left(x + td_k\right) \in C$ as C is closed. So $d \in \mathbf{0}^+ C$. Hence $\mathbf{0}^+ C = C_\infty$.

So if C is non-empty convex, we have $\overline{C}$ is non-empty, closed and convex

$$\mathbf{0}^+ \overline{C} = \left(\overline{C}\right)_\infty = C_\infty$$

by Proposition 2.1. □

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112 原课件第 112 页：衰退锥相等与非闭反例

Proposition of Definition 3

If C is not closed, the theorem may fail. Consider

$$C = \{(x_1, x_2) : 0 \leqq x_1 < 1, x_2 \geqq 0\} \cup \{(1, 0)\}$$

By using the definition of direction of recession, it can be verified that C has no nonzero directions of recession, i.e.

$$\mathbf{0}^+ C = \varnothing$$

but we can see that

$$C_\infty = \mathbf{0}^+ \overline{C} = \{(0, x) : x \in \mathbb{R}\}.$$

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113 原课件第 113 页：衰退锥相等与非闭反例

Proposition of Definition 3

Proposition 4.4. For any non-empty closed convex set $C \subseteq \mathbb{R}^N$, one has $C = C + C_\infty$.

Proof. The inclusion $C \subseteq C + C_\infty$ is clear because $0 \in C_\infty$.

Let $x \in C + C_\infty$. Then there exists $c \in C$, $d \in C_\infty$ with $x = c + d$. Thus, there exists $d_k \in C$, $t_k \to \infty$ such that $\dfrac{d_k}{t_k} \to d$, which implies that for k sufficiently large

$$\left(1 - \frac{1}{t_k}\right) c + \frac{d_k}{t_k} \in C$$

and

$$\left(1 - \frac{1}{t_k}\right) c + \frac{d_k}{t_k} \to c + d = x \in C$$

as C is closed. $\square$

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114 原课件第 114 页：最大可含锥与基点无关

Proposition of Definition 3

Corollary A closed, convex set $C \subseteq \mathbb{R}^N$ owning 0, contains its asymptotic cone.

Proof. As $0 \in C$ and $\{0\} + C_\infty = C_\infty$. As $C = C + C_\infty$, we have

$$C_\infty \subseteq C.$$

□

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闭凸 S、$a \in S$ 下，令 $D_a = \bigcap_{\lambda>0} \lambda(S-a)$。$d \in D_a$ 等价于任意 $t>0$ 有 $a+td \in S$，因此它是从 a 出发可完整放入 S 的最大锥。若 $b \in S$，取 $T \to \infty$，由凸性 $(1-t/T)b + (t/T)(a+Td) \in S$，极限为 $b+td$，闭性给后者在 S，所以基点可以任意换，$D_a = \operatorname{rec} S = S_\infty$。当0在 S 中，取 $a=0, t=1$ 就得到渐近锥包含于原集。本段所有闭性使用均在最后取极限处，不是多余假设。

115 原课件第 115 页：最大可含锥与基点无关

Proposition of Definition 3

Proposition 4.5. A closed, convex set $C \subseteq \mathbb{R}^N$ owning 0, contains its asymptotic cone.

Proof. Let $y \in \mathbf{A}C$. For any $M \geqq 0$, $\mathbb{R}_+ y \in \Gamma\left(C^M\right)$.

(1) $y \in C^M$, then $y \in C$.

(2) $y \notin C^M$, then there exists $\lambda < 1$ such that $\frac{1}{\lambda} y \in C^M \subseteq C$,

hence $y = (1-\lambda)0 + \lambda \frac{1}{\lambda} y \in C$.

Therefore $y \in \mathbf{A}C$ implies $y \in C$. That is, $\mathbf{A}C \subseteq C$. □

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116 原课件第 116 页：最大可含锥与基点无关

Proposition of Definition 3

Proposition 4.6. Let $S \subseteq \mathbb{R}^N$ be a closed convex set. For $x_0 \in S$, define the cone C as follows:

$$C = \max\{C' \mid x_0 + C' \subseteq S\},$$

then C is an asymptotic cone, that is

(1) $C = \bigcap_{\lambda>0} \lambda(S - x_0)$;

(2) C is independent of x_0;

(3) $C = \mathbf{A}S$.

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117 原课件第 117 页：最大可含锥与基点无关

Proposition of Definition 3

Proof of (1). $C \subseteq S - x_0 = S'$ and $0 \in S'$.

$$
\begin{aligned}
y \in C &\Longrightarrow \{\alpha y \mid \alpha \geqq 0\} \subseteq C \\
&\Longrightarrow \{\alpha y \mid \alpha \geqq 0\} \subseteq S' \\
&\Longrightarrow \text{for all } \lambda > 0, \{\alpha y \mid \alpha \geqq 0\} \subseteq \lambda S' \\
&\Longrightarrow \{\alpha y \mid \alpha \geqq 0\} \subseteq \bigcap_{\lambda>0} \lambda S' \\
&\Longrightarrow y \in \bigcap_{\lambda>0} \lambda S'
\end{aligned}
$$

then $C \subseteq \bigcap_{\lambda>0} \lambda(S - x_0)$. Conversely,

$$
\begin{aligned}
y \in \bigcap_{\lambda>0} \lambda S' &\Longrightarrow \{\alpha y \mid \alpha \geqq 0\} \subseteq \bigcap_{\lambda>0} \lambda S' \\
&\Longrightarrow \text{for all } \lambda > 0, \{\alpha y \mid \alpha \geqq 0\} \subseteq \lambda S' \\
&\Longrightarrow \{\alpha y \mid \alpha \geqq 0\} \subseteq S' \\
&\Longrightarrow \{\alpha y \mid \alpha \geqq 0\} \subseteq C \\
&\Longrightarrow y \in C
\end{aligned}
$$

then $\bigcap_{\lambda>0} \lambda(S - x_0) \subseteq C$. $\square$

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118 原课件第 118 页：最大可含锥与基点无关

Proposition of Definition 3

Proof of (2). Suppose $x_0 \in S$, $y \in S \Longrightarrow \mathbb{R}_+ y \subseteq C \subseteq S'$. For any $x \in S$,

$$x_0 + \mathbb{R}_+ y \subseteq S \Longleftrightarrow x + \mathbb{R}_+ y \subseteq S$$

then C is independent of x_0.

Proof of (3).

$$\mathbb{R}_+ y \subseteq \bigcap_{\lambda>0} \lambda S' \Longrightarrow \text{for any } \lambda > 0, \mathbb{R}_+ y \subseteq \lambda S' \Longrightarrow \mathbb{R}_+ y \in \mathbf{A}S' = \mathbf{A}S$$

then $\bigcap_{\lambda>0} \lambda S' \subseteq \mathbf{A}S$. Conversely,

$$\begin{aligned} \mathbb{R}_+ y \in \mathbf{A}S = \mathbf{A}S' &\Longrightarrow \text{for any } M > 0, \mathbb{R}_+ y \subseteq \Gamma(S'^M) \\ &\Longrightarrow \text{for any } M > 0, [\alpha_M, \infty) y \subseteq S'^M \\ &\Longrightarrow \mathbb{R}_+ y \subseteq S' \\ &\Longrightarrow \text{for any } \lambda > 0, \mathbb{R}_+ y \subseteq \lambda S' \\ &\Longrightarrow \mathbb{R}_+ y \subseteq \bigcap_{\lambda>0} \lambda S' \end{aligned}$$

then $\mathbf{A}S \subseteq \bigcap_{\lambda>0} \lambda S'$. $\square$

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